

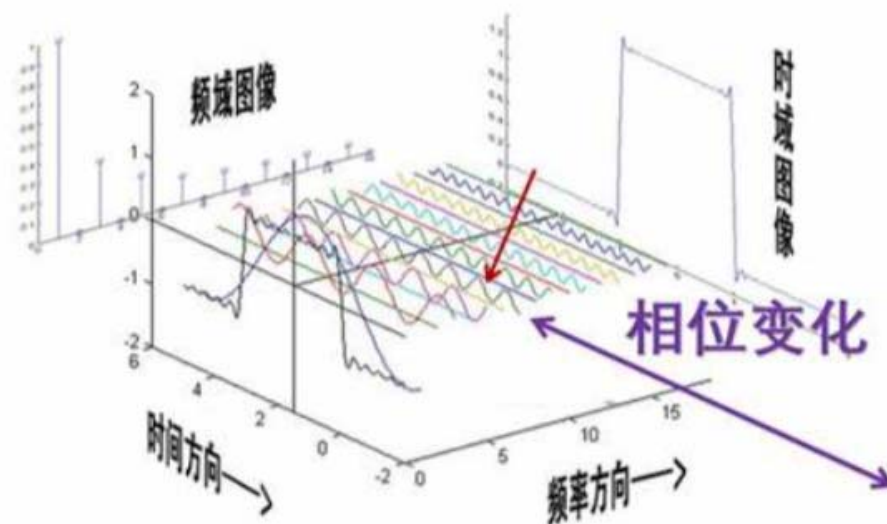
第4章 频率域滤波

换个角度看问题，生命会展现出另一种美。
生活中不是缺少美，而是缺少发现。

——罗丹

内容提纲

- 基本概念
- 取样和取样函数的傅里叶变换
- 离散傅里叶变换
- 频率域滤波基础
- 利用频率域滤波平滑和锐化图像
- 实现



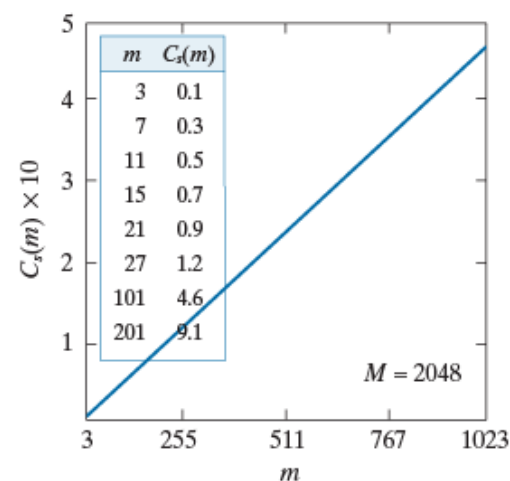
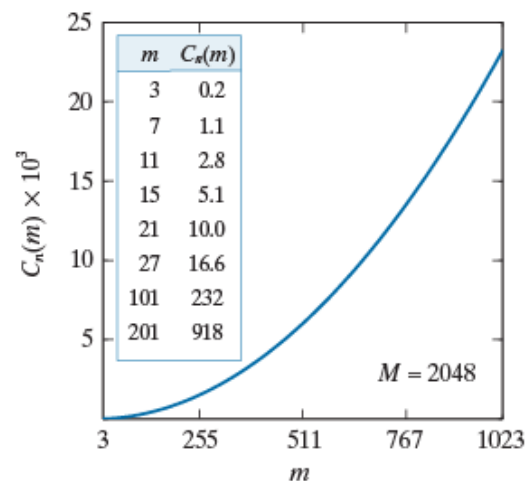
频率域与空域滤波的效率对比

- 采用不可分离模板时，采用FFT滤波相比空域滤波的计算优势为：

$$\begin{aligned} C_n(m) &= \frac{M^2 m^2}{2M^2 \log_2 M^2} \\ &= \frac{m^2}{4 \log_2 M} \end{aligned}$$

- 采用可分离模板时，采用FFT滤波相比空域滤波的计算优势为：

$$\begin{aligned} C_s(m) &= \frac{2M^2 m}{2M^2 \log_2 M^2} \\ &= \frac{m}{2 \log_2 M} \end{aligned}$$



- 不可分离模板（左）与可分离模板（右）

傅里叶级数和傅里叶变换

- **傅里叶级数**: 任何(无限)周期函数可以表示为不同频率的正弦和/或余弦之和的形式;
- **傅里叶变换**: 非周期函数也可以用正弦和/或余弦乘以加权函数的积分来表示;
- **最重要的性质**: 用傅里叶变换表示的函数特征完全可以通过傅里叶反变换来重建, 而不会丢失任何信息。

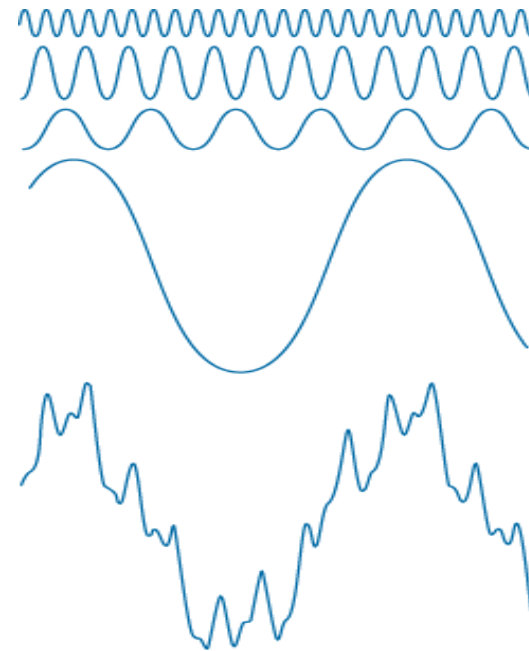


FIGURE 4.1
The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

复数

- 复数 C 定义为 $C = R + jI$, 其中 $j = \sqrt{-1}$;
- 复数 C 共轭表示为 $C^* = R - jI$;
- 也可以用极坐标来表示复数: $C = |C|(\cos \theta + j \sin \theta)$
- 使用欧拉公式 $e^{j\theta} = \cos \theta + j \sin \theta$, 其中 $e = 2.71828\cdots$, 则有极坐标下复数表示:

$$C = |C|e^{j\theta}$$

傅里叶级数

- 具有周期 T 的连续变量 t 的周期函数 $f(t)$ 可以被描述为乘以适当系数的正弦和余弦和。

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\frac{2\pi n}{T}t}$$
$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-j\frac{2\pi n}{T}t} dt, n = 0, \pm 1, \pm 2, \dots$$

冲激及其取样特性

- 连续变量的单位冲激定义：
$$\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases}$$

- 限制条件：
$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

- 冲激具有的取样特性

$$\int_{-\infty}^{\infty} f(t) \delta(t) dt = f(0)$$

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0)$$

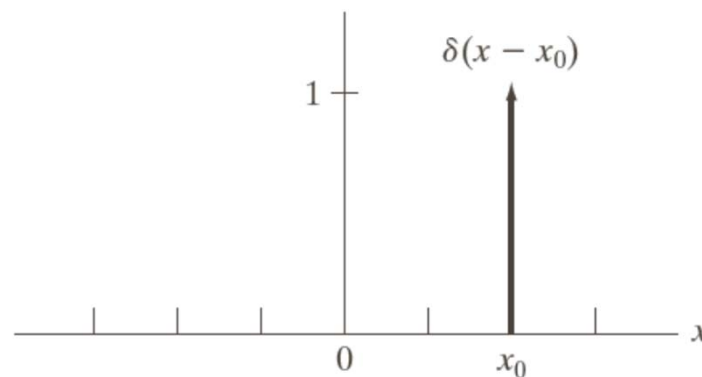
单位离散冲激

- 类似地，定义单位离散冲激

$$\delta(x) = \begin{cases} 1, & x = 0 \\ 0, & x \neq 0 \end{cases}$$

- 同样满足 $\sum_{x=-\infty}^{\infty} \delta(x) = 1$

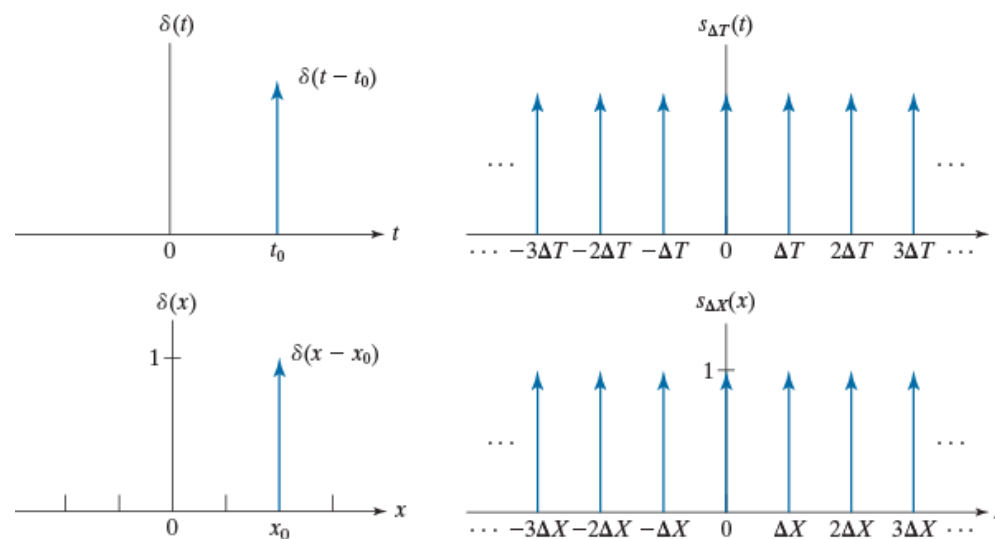
- 取样特性 $\sum_{x=-\infty}^{\infty} f(x)\delta(x) = f(0)$, 以及 $\sum_{x=-\infty}^{\infty} f(x)\delta(x-x_0) = f(x_0)$



冲激串

- 定义：
 - 无限多个分离的周期冲激单元之和

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} \delta(t - n\Delta T)$$



上：连续单位冲激与连续单位冲激串
下：离散单位冲激与离散单位冲激串

连续变量函数的傅里叶变换

- 定义正变换

$$F(u) = F\{f(t)\} = \int_{-\infty}^{\infty} f(t) e^{-j2\pi ut} dt$$

- 反变换

$$f(t) = \int_{-\infty}^{\infty} F(u) e^{j2\pi ut} du$$

- 傅里叶变换域即频率域；
- 如果 $f(t)$ 是实数，其变换通常是复数；
- t 可以表示任何连续变量，例如时域的“秒”，空域的“米”等。 u 呢？

简单窗函数的傅里叶变换

- $F(u)$ 的零值位置与“盒状”函数的宽度 W 成反比;
- $F(u)$ 旁瓣高度随距离原点的增大而减少;
- $F(u)$ 函数向横轴正负方向无限扩展

$$\begin{aligned} F(u) &= \int_{-\infty}^{\infty} f(t) e^{-j2\pi ut} dt = \int_{-W/2}^{W/2} A e^{-j2\pi ut} dt \\ &= \frac{-A}{j2\pi u} \left[e^{-j2\pi ut} \right]_{-W/2}^{W/2} = AW \frac{\sin(\pi u W)}{\pi u W} \end{aligned}$$

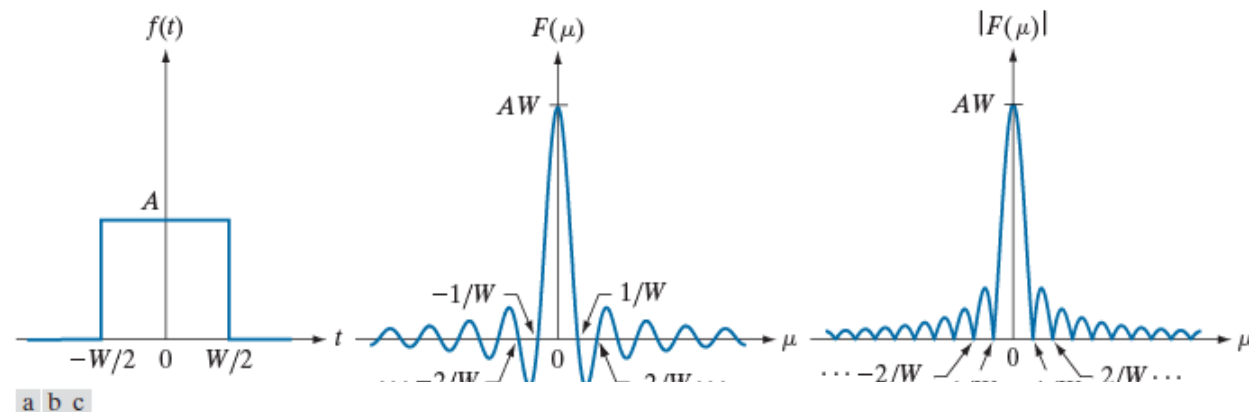


FIGURE 4.4 (a) A box function, (b) its Fourier transform, and (c) its spectrum. All functions extend to infinity in both directions. Note the inverse relationship between the width, W , of the function and the zeros of the transform.

冲激的傅里叶变换

- 位于空间域原点的冲激的傅里叶变换是一个常数。

$$F(u) = \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi ut} dt = \int_{-\infty}^{\infty} e^{-j2\pi ut} \delta(t) dt = e^{-j2\pi u0} = 1$$

- 位于 $t = t_0$ 的一个冲激的傅里叶变换是：

$$\begin{aligned} F(u) &= \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j2\pi ut} dt = \int_{-\infty}^{\infty} e^{-j2\pi ut} \delta(t - t_0) dt \\ &= e^{-j2\pi ut_0} = \cos(2\pi ut_0) - j\sin(2\pi ut_0) \end{aligned}$$

空域平移对应于频域相移

变换对特性

- 由傅里叶反变换公式：
$$f(t) = \int_{-\infty}^{\infty} F(u) e^{j2\pi ut} du$$
- 有：
$$\begin{aligned} \int_{-\infty}^{\infty} F(u-a) e^{j2\pi ut} du &= \int_{-\infty}^{\infty} F(u-a) e^{j2\pi(u-a)t} e^{j2\pi at} d(u-a) \\ &= e^{j2\pi at} \int_{-\infty}^{\infty} F(u-a) e^{j2\pi(u-a)t} d(u-a) \\ &= e^{j2\pi at} f(t) \end{aligned}$$

- 即若 $f(t) \Leftrightarrow F(u)$, 则有 $f(t)e^{j2\pi at} \Leftrightarrow F(u-a)$

- 同理, 有:

$$\delta(t-t_0) \Leftrightarrow e^{-j2\pi ut_0}$$

$$e^{j2\pi t_0} \Leftrightarrow \delta(u-t_0)$$

频域平移对应于空域相移

冲激串的傅里叶变换

- 冲激串的傅里叶级数表示 $s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\frac{2\pi n}{\Delta T}t} = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t}$
- 其傅里叶变换:

$$\begin{aligned} S(u) &= F\{s_{\Delta T}(t)\} = F\left\{\frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t}\right\} \\ &= \frac{1}{\Delta T} F\left\{\sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t}\right\} = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \left(u - \frac{n}{\Delta T}\right) \end{aligned}$$

- 周期为 ΔT 的冲激串的傅里叶变换还是冲激串，其周期为 $\frac{1}{\Delta T}$ （频谱周期延拓）

卷积定理

- 函数卷积定义

$$f(t) * h(t) = \int_{-\infty}^{\infty} f(\tau)h(t-\tau)d\tau$$

- 卷积定理

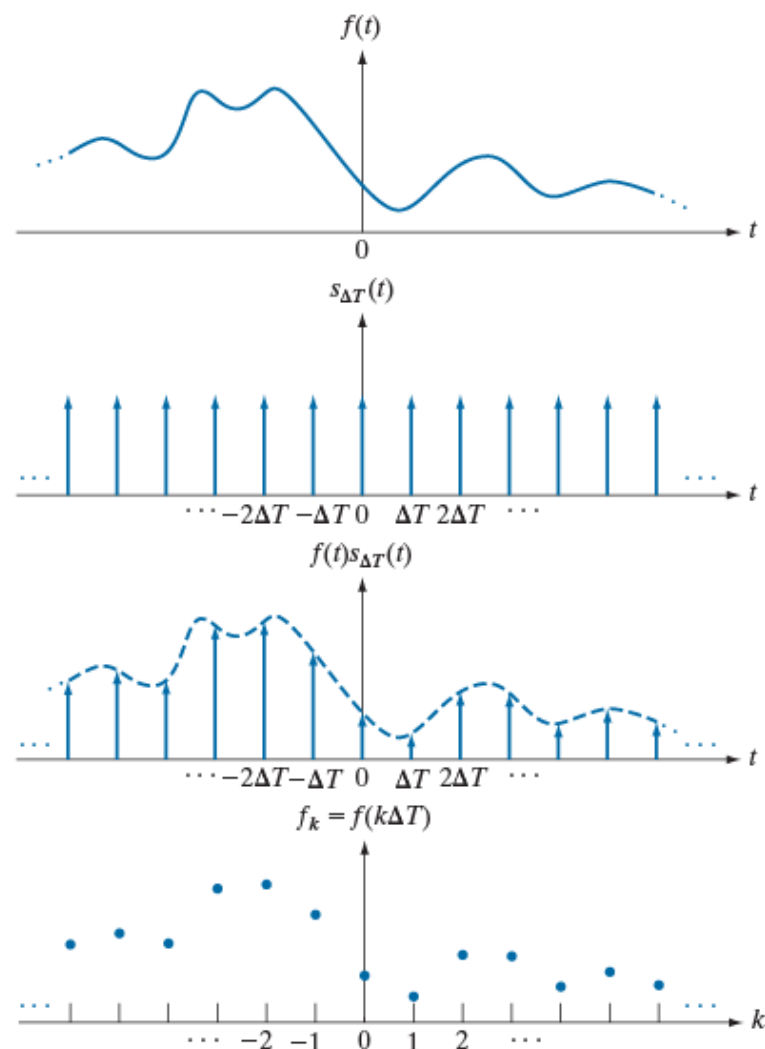
$$f(t) * h(t) \Leftrightarrow H(u)F(u)$$

$$f(t)h(t) \Leftrightarrow H(u) * F(u)$$

信号取样

- 取样后的函数

$$\tilde{f}(t) = f(t)s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} f(t)\delta(t - n\Delta T)$$

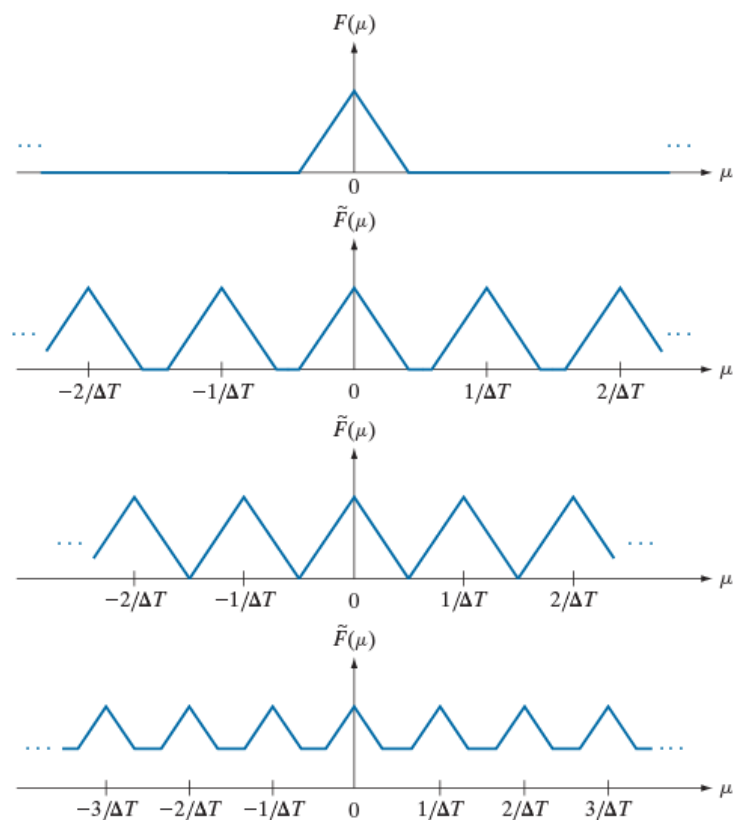


函数被取样后的傅里叶变换

- 对应于各自傅里叶变换的卷积

$$\begin{aligned}\tilde{F}(u) &= F(u) * S(u) = \int_{-\infty}^{\infty} F(\tau) S(u - \tau) d\tau \\ &= \frac{1}{\Delta T} \int_{-\infty}^{\infty} F(\tau) \sum_{n=-\infty}^{\infty} \delta(u - \tau - \frac{n}{\Delta T}) d\tau \\ &= \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} F(u - \frac{n}{\Delta T})\end{aligned}$$

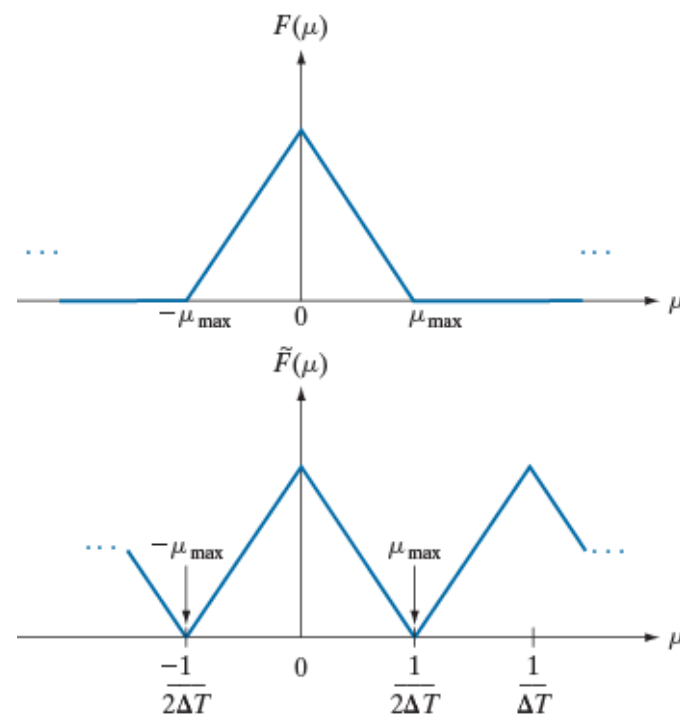
- $\tilde{F}(u)$ 是连续的，由原信号的周期拷贝构成（虽然 $\tilde{f}(t)$ 是离散的）



取样定理

- 对于以原点为中心的有限区间 $[-u_{\max}, u_{\max}]$ 之外的频率值，其傅里叶变换为零的函数 $f(t)$ 称为带限函数。
- 那奎斯特取样定理： $\frac{1}{\Delta T} > 2u_{\max}$
- 思考：1. $\frac{1}{\Delta T} = 2u_{\max}$ 可以吗？不可以。
- 2. 怎么样的函数是带限的？

$f(t)$ 必须从 $-\infty$ 到 $+\infty$



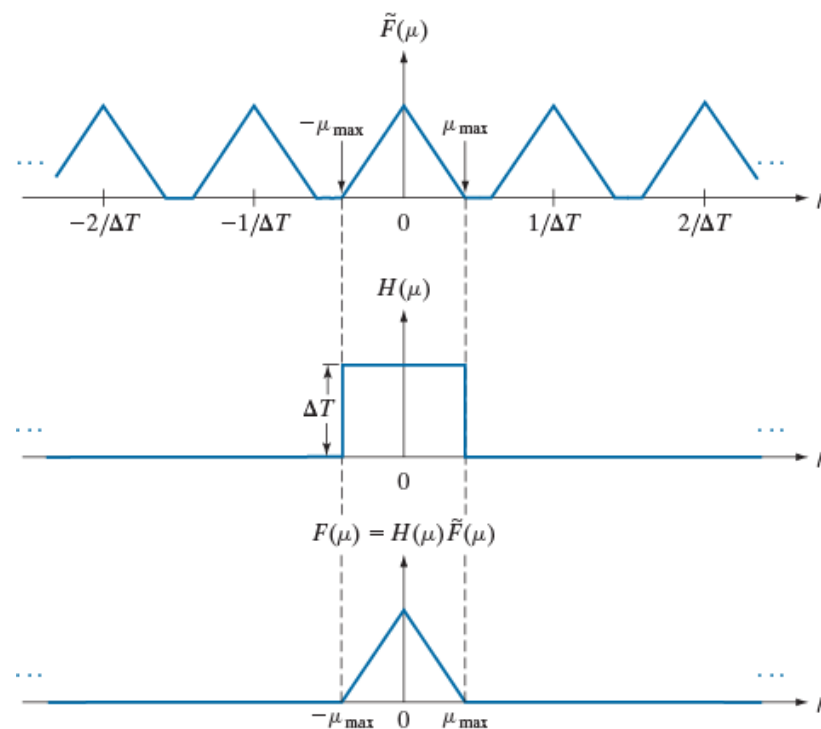
重建滤波器与恢复

- 重建滤波器

$$H(u) = \begin{cases} \Delta T, & -u_{\max} \leq u \leq u_{\max} \\ 0, & \text{其他} \end{cases}$$

- 恢复

$$F(u) = H(u)\tilde{F}(u)$$



频谱混淆

- 在实践中，混淆总是存在的。在我们限制函数的持续时间时，**相当于对原信号加窗**，必然引进无限的频率分量。
- 有限持续时间的函数都不是带限的**，反过来，一个带限函数一定从负无穷到正无穷。
- 在实践中，通过平滑输入函数降低混淆。它**必须在函数被取样之前完成**。

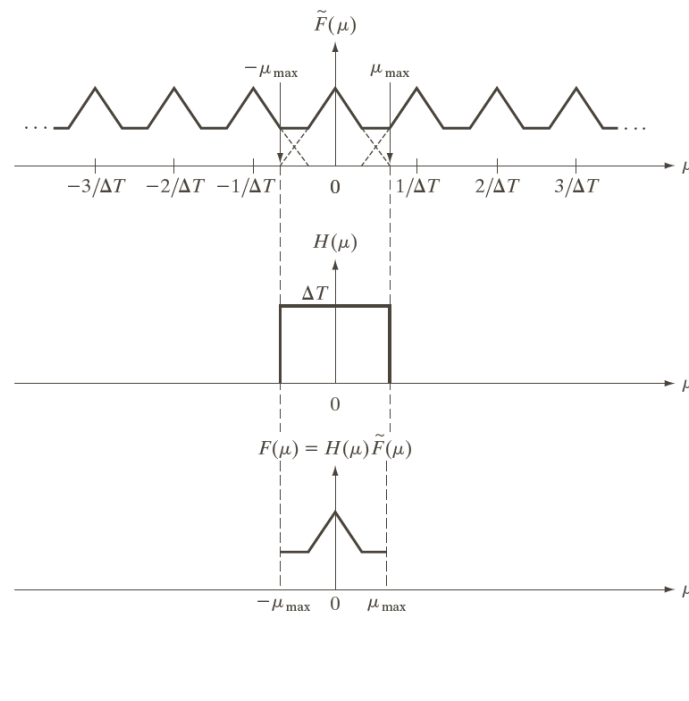
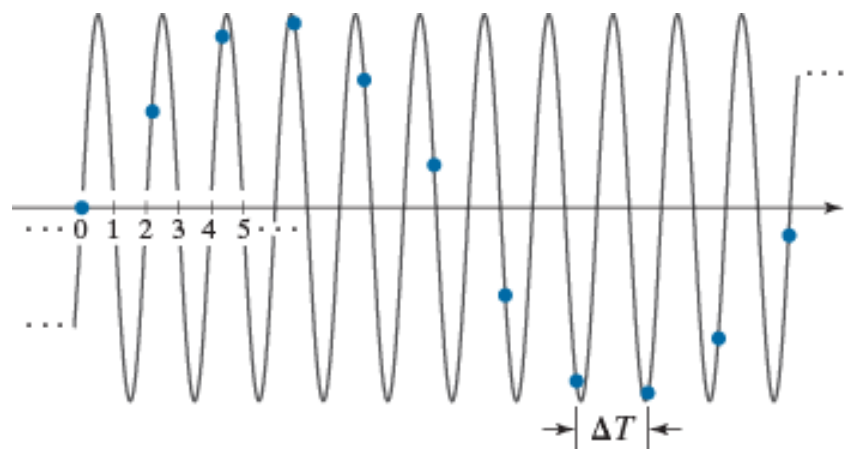
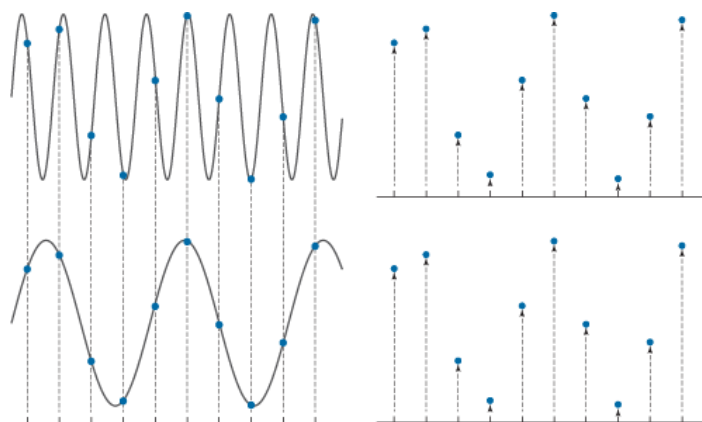


FIGURE 4.9 (a) Fourier transform of an under-sampled, band-limited function. (Interference from adjacent periods is shown dashed in this figure). (b) The same ideal lowpass filter used in Fig. 4.8(b). (c) The product of (a) and (b). The interference from adjacent periods results in aliasing that prevents perfect recovery of $F(\mu)$ and, therefore, of the original, band-limited continuous function. Compare with Fig. 4.8.

混淆的例子

- 取样率过低，得到的是另外一个正弦信号
- 取样要大于两倍原始信号频率，刚好两倍也不行。如正弦信号的零点。



由取样后的数据重建函数

- 由于 $F(u) = H(u)\tilde{F}(u)$
- 有：
$$f(t) = F^{-1}\{F(u)\} = F^{-1}\{H(u)\tilde{F}(u)\} = h(t) * \tilde{f}(t)$$
- 即：
$$f(t) = \sum_{-\infty}^{\infty} f(n\Delta T) \text{sinc}[(t - n\Delta T) / \Delta T]$$
- Sinc(0)=1,且对于任何其他整数m, sinc(m)=0,
- 样本点的值就是原采样值, 而样本点之间的值由sinc函数形成内插

离散傅里叶变换

- 初衷：实践中，空域和频域都只能处理有限数量的样本。
- 对 $0 \leq u \leq 1/\Delta T$ 一个周期内取样 M 个样本，取样位置为

$$u = \frac{m}{M\Delta T}, \quad m = 0, 1, 2, \dots, M-1$$

- 得到离散傅里叶变换对

$$F(u) = \sum_{x=0}^{M-1} f(x)e^{-j2\pi ux/M}, \quad u = 0, 1, 2, \dots, M-1$$

$$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u)e^{j2\pi ux/M}, \quad x = 0, 1, 2, \dots, M-1$$

容易证明，正反傅里叶变换都是无限的，周期的，周期为 M 。

取样和频率间隔之间的关系

- 空域由M个间隔为 ΔT 的样本组成，则持续时间为 $T = M \Delta T$
- 离散频率域的间隔为 $\Delta u = \frac{1}{M \Delta T} = \frac{1}{T}$
- 由DFT的M个分量跨越的整个频率范围是 $\Omega = M \Delta u = \frac{1}{\Delta T}$
- DFT的频率分辨率取决于空域持续时间，跨越的频率范围取决于采样间隔。（一个域中的间隔与另一个域中的范围之间的反比关系）
- 思考：能否同时获得空域与频域的高分辨率？
 - 空域持续时间足够长，采样间隔足够小->频域分辨率足够高，频率范围足够大。
 - 无限接近于完全连续状态

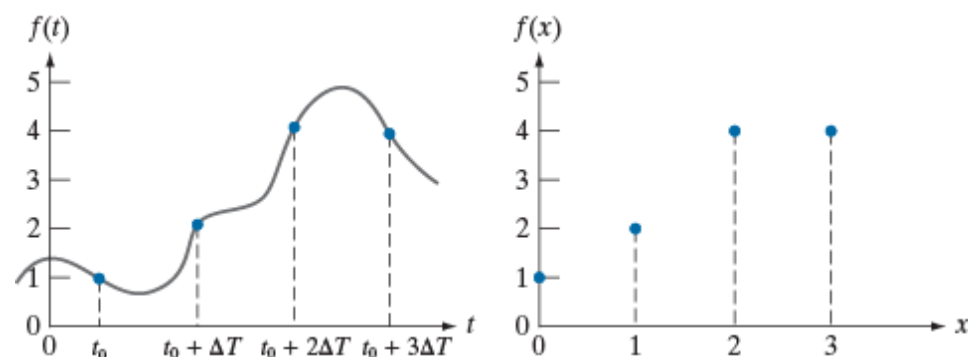
例子：计算DFT

- 正变换：

$$F(0) = \sum_{x=0}^3 f(x) = [f(0) + f(1) + f(2) + f(3)] = 1 + 2 + 4 + 4 = 11$$

$$F(1) = \sum_{x=0}^3 f(x) e^{-j2\pi(1)x/4} = 1e^0 + 2e^{-j\pi/2} + 4e^{-j\pi} + 4e^{-j3\pi/2} = -3 + 2j$$

$$F(2) = -(1 + 0j) \text{ and } F(3) = -(3 + 2j).$$



- 反变换：

$$f(0) = \frac{1}{4} \sum_{u=0}^3 F(u) e^{j2\pi u(0)} = \frac{1}{4} \sum_{u=0}^3 F(u) = \frac{1}{4} [11 - 3 + 2j - 1 - 3 - 2j] = \frac{1}{4} [4] = 1$$

4.5 二维情况的扩展 (1)

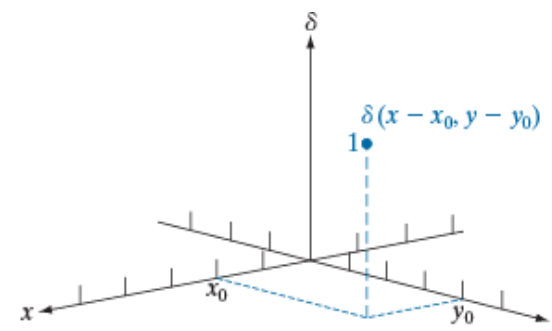
- 二维连续变量的单位冲激定义: $\delta(t, z) = \begin{cases} \infty, & t = z = 0 \\ 0, & \text{其他} \end{cases}$

- 限制条件: $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(t, z) dt dz = 1$

- 冲激具有的取样特性

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) \delta(t, z) dt dz = f(0, 0)$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) \delta(t - t_0, z - z_0) dt dz = f(t_0, z_0)$$



二维情况的扩展 (2)

- 二维连续变量正变换

$$F(u,v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t,z) e^{-j2\pi(ut+vz)} dt dz$$

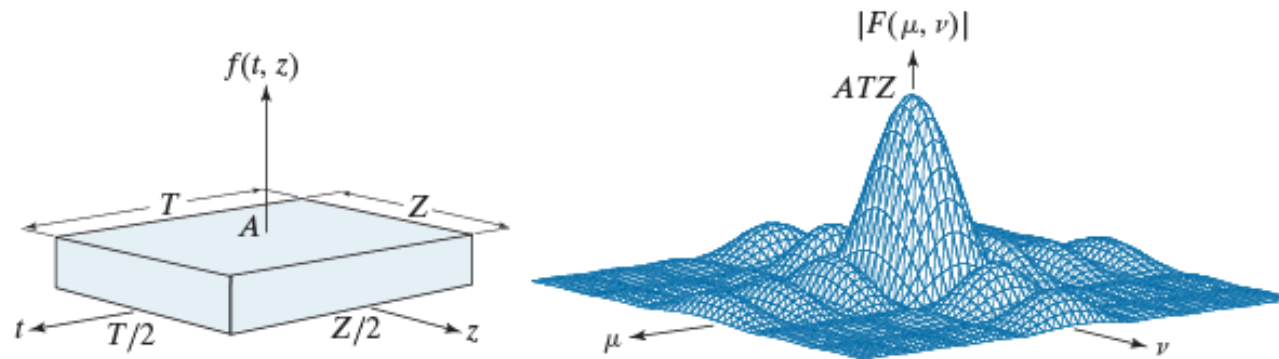
- 反变换

$$f(t,z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u,v) e^{j2\pi(ut+vz)} du dv$$

矩形窗函数的二维傅里叶变换

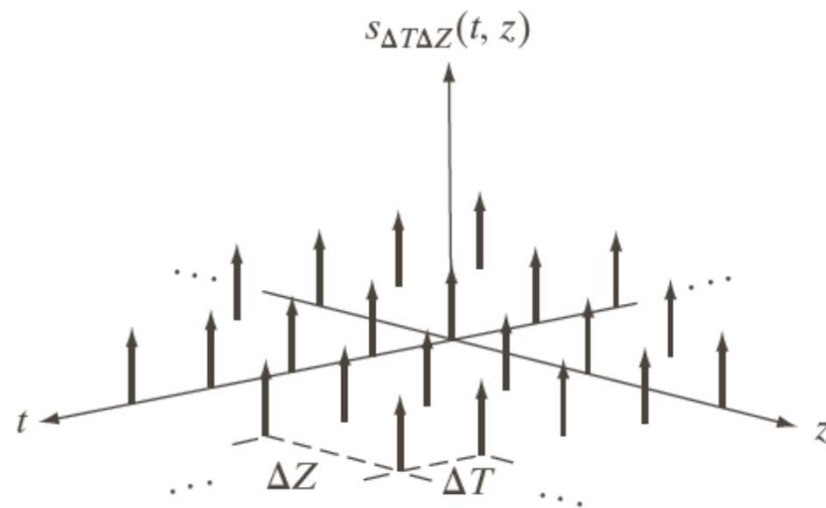
- 二维窗函数及其傅里叶变换

$$\begin{aligned} F(u, v) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) e^{-j2\pi(ut+vz)} dt dz = \int_{-T/2}^{T/2} \int_{-Z/2}^{Z/2} A e^{-j2\pi(ut+vz)} dt dz \\ &= ATZ \frac{\sin(\pi u T)}{\pi u T} \frac{\sin(\pi v Z)}{\pi v Z} \end{aligned}$$



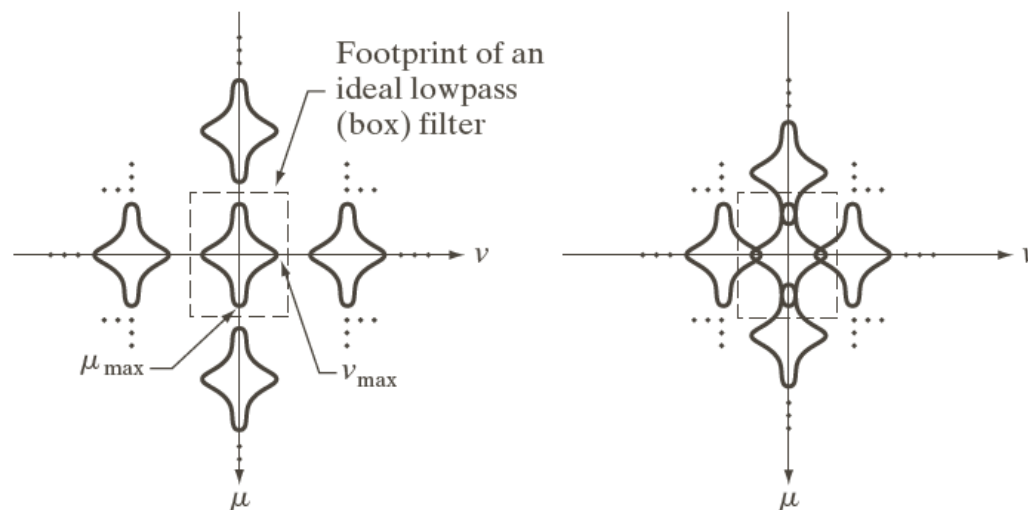
二维取样和取样定理

$$s_{\Delta T \Delta Z}(t, z) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(t - m\Delta T, z - n\Delta Z)$$



二维取样定理

- 若有 $\frac{1}{\Delta T} > 2u_{\max}$ 和 $\frac{1}{\Delta Z} > 2v_{\max}$
- 则原连续带限函数可以由其一组样本无误地恢复。



图像中的混淆

- 图像中存在两种主要的混淆现象：**空间混淆**和**时间混淆**。
- **时间混淆**的一个例子是“车轮”效应。即在序列图像中具有轮辐的车轮出现倒转的现象。这是由于序列的帧率低于轮子转动的速度引起的。



空间混淆的例子

- 用96*96的像素方阵去对不同尺寸黑白格子采样成像后，获得的图像。
 - 左上格子边长：16像素；
 - 右上格子边长：6像素；
 - 左下格子边长：0.95像素；
 - 右下格子边长：0.5像素。
- 注意最后一幅看起来像完全正常的图像。

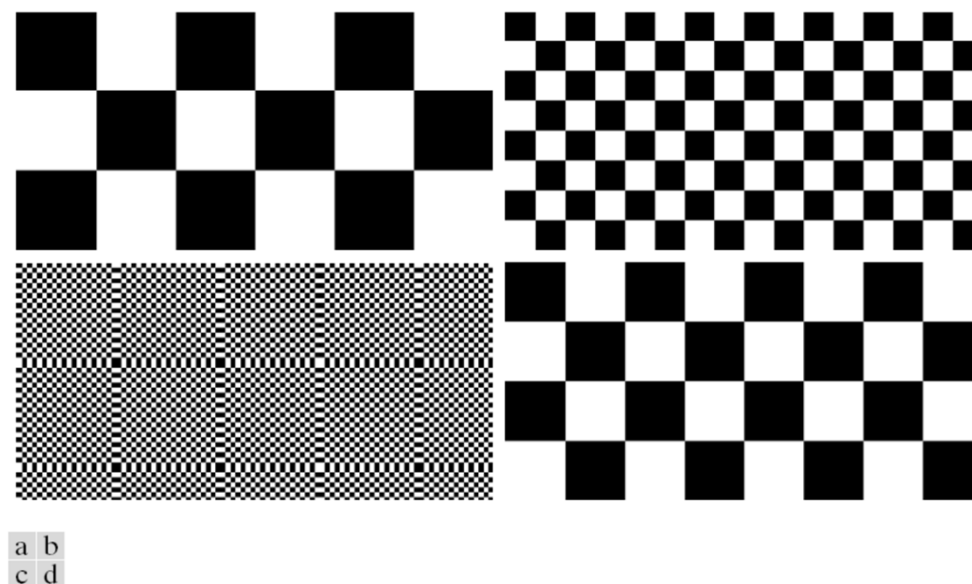
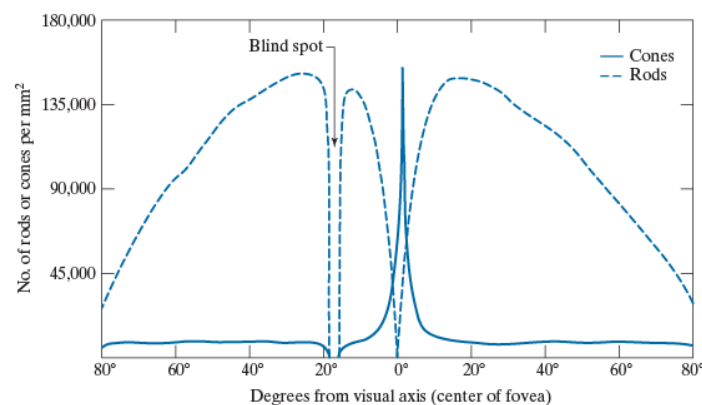
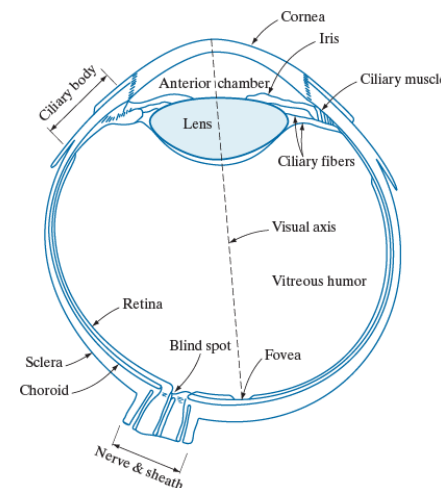


FIGURE 4.16 Aliasing in images. In (a) and (b), the lengths of the sides of the squares are 16 and 6 pixels, respectively, and aliasing is visually negligible. In (c) and (d), the sides of the squares are 0.9174 and 0.4798 pixels, respectively, and the results show significant aliasing. Note that (d) masquerades as a “normal” image.

人眼视网膜的采样率

- 测试表明黑白条带超过120条/度（60周期/度）时，观察者将图像感知为一个灰度方块；
- 根据采样定理推测，人眼有120个锥状体/度，事实证明人眼确实如此。



重取样后图像中的混淆例子

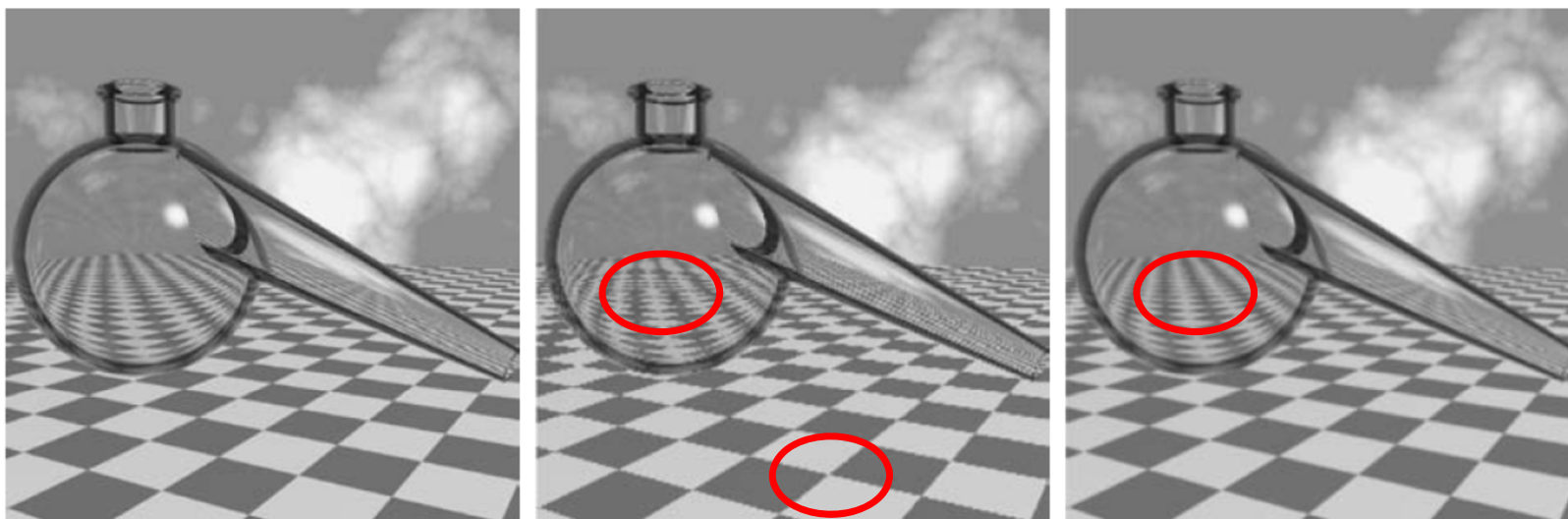
- 1 原图
- 2 降采样50%后重放大
- 3 先3*3均值平滑滤波后再降采样50%，后重放大



a b c

FIGURE 4.17 Illustration of aliasing on resampled images. (a) A digital image with negligible visual aliasing. (b) Result of resizing the image to 50% of its original size by pixel deletion. Aliasing is clearly visible. (c) Result of blurring the image in (a) with a 3×3 averaging filter prior to resizing. The image is slightly more blurred than (b), but aliasing is not longer objectionable. (Original image courtesy of the Signal Compression Laboratory, University of California, Santa Barbara.)

图像降采样后方法带来的锯齿现象（欠取样）

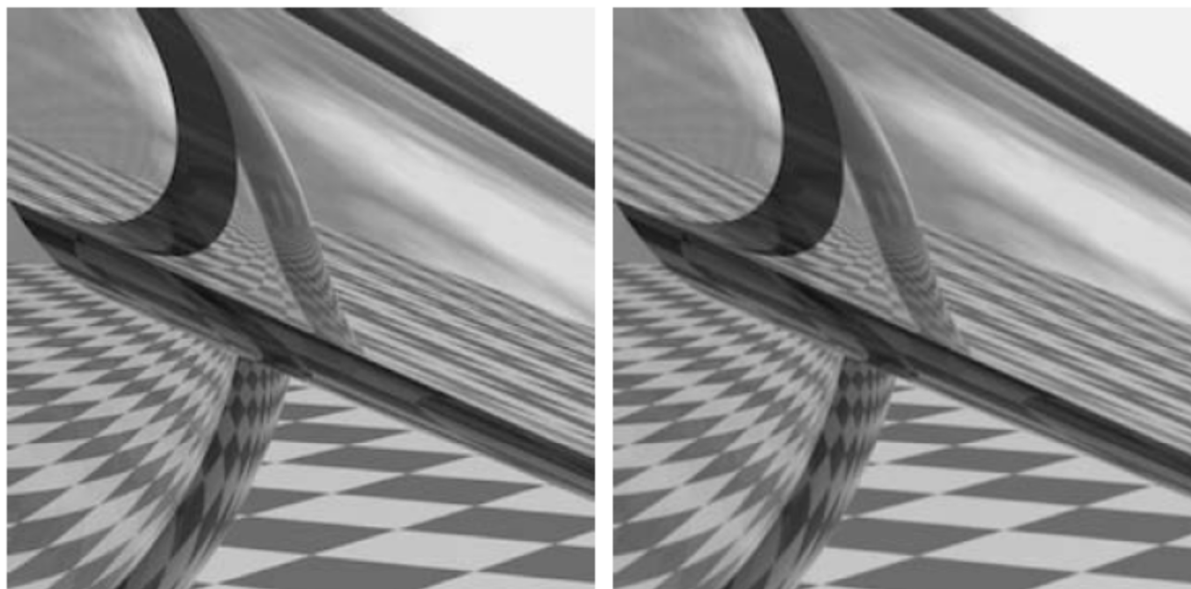


a b c

FIGURE 4.18 Illustration of jaggies. (a) A 1024×1024 digital image of a computer-generated scene with negligible visible aliasing. (b) Result of reducing (a) to 25% of its original size using bilinear interpolation. (c) Result of blurring the image in (a) with a 5×5 averaging filter prior to resizing it to 25% using bilinear interpolation. (Original image courtesy of D. P. Mitchell, Mental Landscape, LLC.)

图像放大中锯齿现象

1. 像素复制法放大
2. 双线性内插法放大

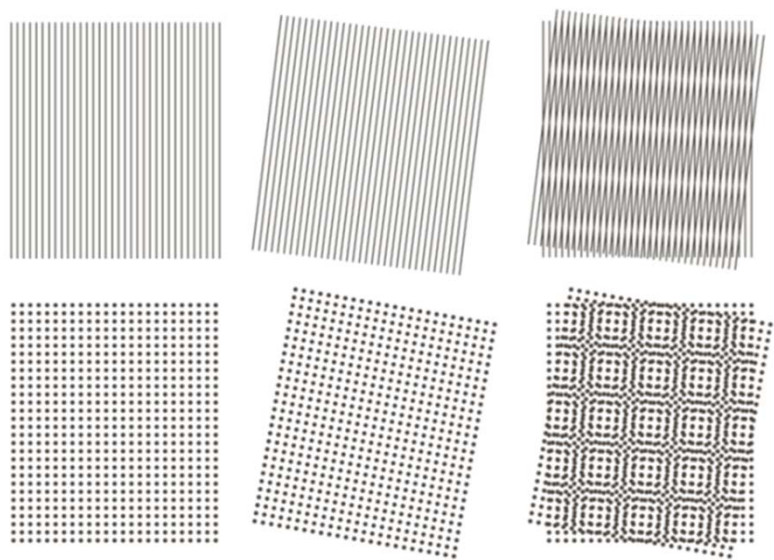


a b

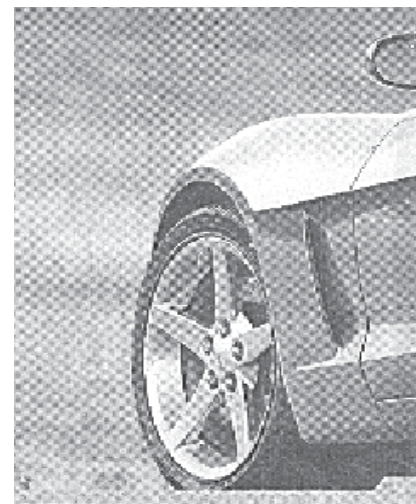
FIGURE 4.19 Image zooming. (a) A 1024×1024 digital image generated by pixel replication from a 256×256 image extracted from the middle of Fig. 4.18(a). (b) Image generated using bi-linear interpolation, showing a significant reduction in jaggies.

莫尔模式

- 用周期或者近似周期分量对场景取样产生的。
- 将一个（取样）模式叠加在另外一个（场景）模式上，等价于两个模式相乘



数字化取样点阵（纵横）和报纸图像上的点模式（正负**45**度方向）相互作用，产生的均匀莫尔模式。



二维离散傅里叶变换

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)},$$

$$u = 0, 1, 2, \dots, M-1, \quad v = 0, 1, 2, \dots, N-1$$

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)},$$

$$x = 0, 1, 2, \dots, M-1, \quad y = 0, 1, 2, \dots, N-1$$

二维离散傅里叶变换的性质

- 1. 空间和频率间隔的关系

$$\Delta u = \frac{1}{M \Delta T}$$

$$\Delta v = \frac{1}{N \Delta Z}$$

- 频率域样本的间隔与空间样本的持续时间成反比

- **2. 平移性**

- 因为 $f(x - x_0, y - y_0) \Leftrightarrow F(u, v) \exp[-j2\pi(ux_0 / M + vy_0 / N)]$

- 和 $f(x, y) \exp[j2\pi(u_0x / M + v_0y / N)] \Leftrightarrow F(u - u_0, v - v_0)$

- 空间域平移 ($\mathbf{x}_0, \mathbf{y}_0$) 等价于在频域中线性相位改变, 但幅度不变

- 频域中平移 ($\mathbf{u}_0, \mathbf{v}_0$), 等价于在空间域被一复正弦函数相乘, 也相当于在空间域被调制到 ($\mathbf{u}_0, \mathbf{v}_0$) 频率上

- **3. 旋转不变性**

- 先转换到极坐标 : $x = r \cos \theta, \quad y = r \sin \theta$
- 图像表示为 : $f(r, \theta)$
- 变换域也用极坐标: $u = w \cos \phi, \quad v = w \sin \phi$
- 变换域表示成 : $F(w, \phi)$
- 则有 $f(r, \theta + \theta_0) \Leftrightarrow F(w, \phi + \theta_0)$

- **4. 分配律**

$$F\{f_1(x, y) + f_2(x, y)\} = F\{f_1(x, y)\} + F\{f_2(x, y)\}$$

- 对加法满足分配律，但对乘法不满足分配律

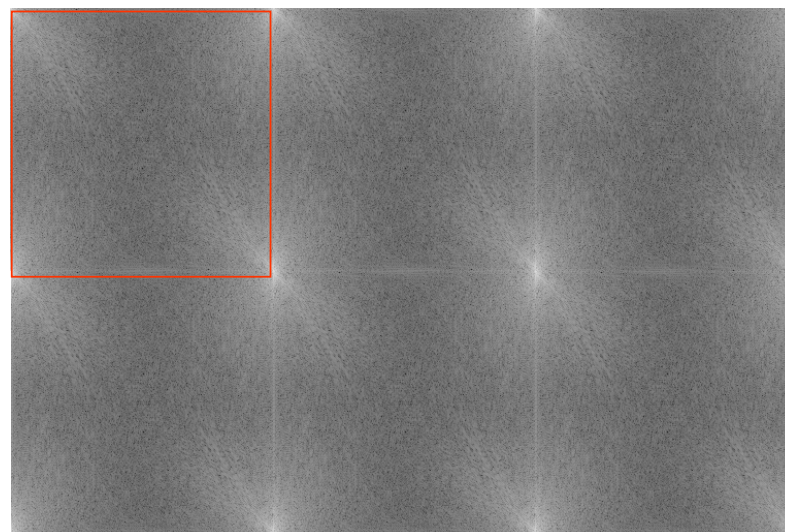
- 5. 周期性

- 若 M, N 是长度，用 $u+mM$ 代 u ，用 $v+nN$ 代 v ，其中 $m, n=0, \pm 1, \pm 2, \dots$ 则有

$$F(u+mM, v+nN) = \frac{1}{N} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \exp[-j2\pi(ux/M + vy/N)] \cdot \exp[-j2\pi(mx + ny)] = F(u, v)$$

用到 $\exp[-j2\pi(mx + ny)] = 1$

- 从频域变换回来的空间域也有这样的周期性



周期性循环的二维频谱

周期性与中心点平移

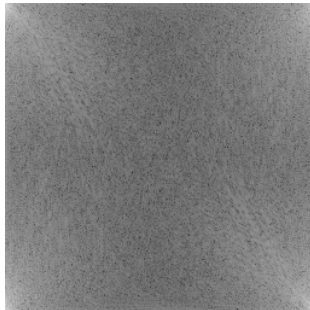
- 特别地, 若 $u_0 = \frac{M}{2}, v_0 = \frac{N}{2}$
- 有

$$\exp[j2\pi(u_0x/M + v_0y/N)] = \exp[j\pi(x+y)] = (-1)^{x+y}$$

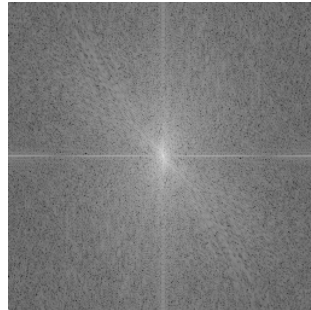
- 故

$$f(x, y)(-1)^{x+y} \Leftrightarrow F(u - \frac{M}{2}, v - \frac{N}{2})$$

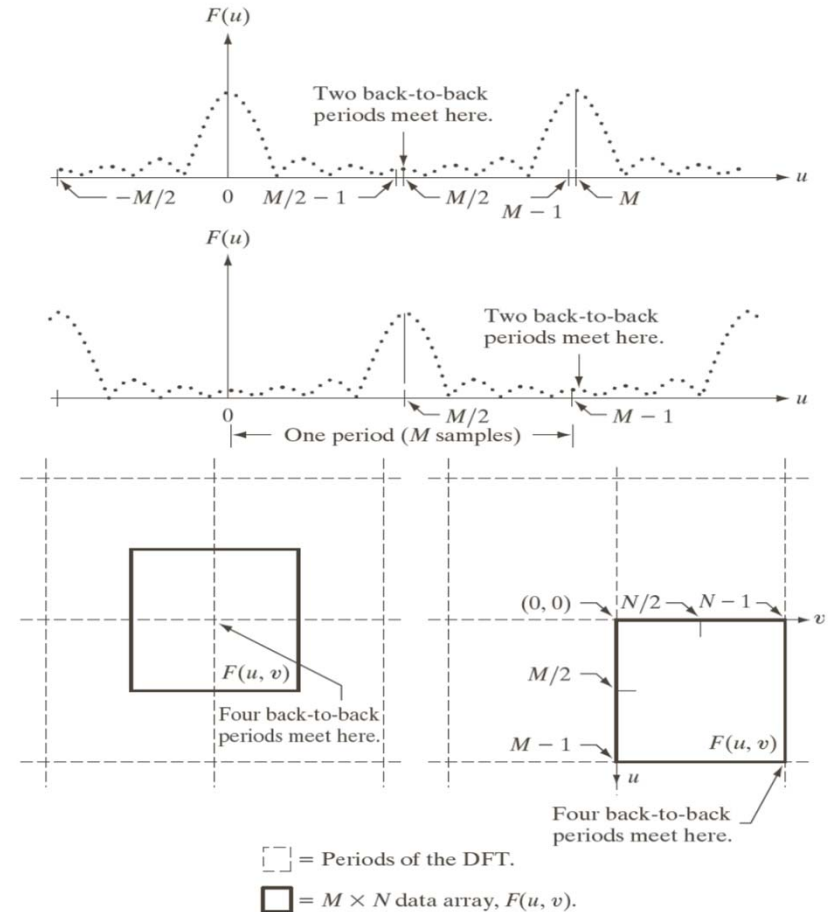
$$f(x, y)(-1)^{x+y} \Leftrightarrow F(u + \frac{M}{2}, v + \frac{N}{2})$$



Lenna图的频谱图



原点移到中心的频谱图



6. 对称性

- 任意实函数或者虚函数都可以表示成为一个奇数部分和偶数部分的和：

$$w(x, y) = w_e(x, y) + w_o(x, y) \quad \text{其中:} \quad w_e(x, y) = \frac{w(x, y) + w(-x, -y)}{2}$$
$$w_o(x, y) = \frac{w(x, y) - w(-x, -y)}{2}$$

- 在DFT和IDFT中，对称性指关于序列中点的对称，即：

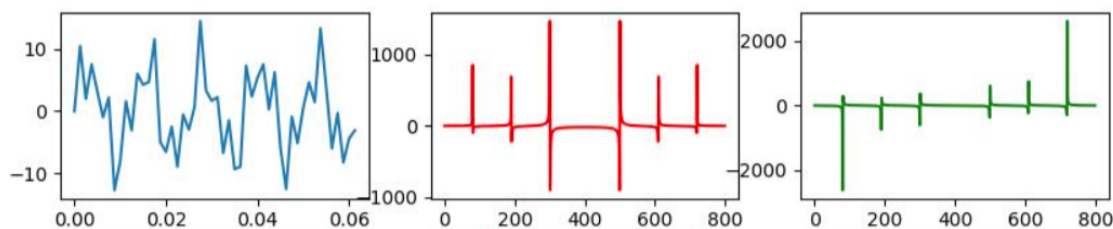
$$w_e(x, y) = w_e(M-x, M-y)$$

$$w_o(x, y) = -w_o(M-x, M-y)$$

对称性

- 对任意一个离散偶函数和奇函数，有：

$$\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} w_e(x, y) w_o(x, y) = 0$$



实函数的傅里叶变换是共轭对称的

- 实函数的傅里叶变换是共轭对称的，即：

$$F^*(u, v) = F(-u, -v)$$

- 虚函数的傅里叶变换是共轭反对称的：

$$F^*(u, v) = -F(-u, -v)$$

偶函数与奇函数例子

- 例1: $f=\{2,1,1,1\}$
 - 偶函数, 关于 $f(2)$ 对称, $f(0)$ 不影响

- 例2: $f=\{0,-1,0,1\}$
 - 奇函数, 关于 $f(2)$ 奇对称, $f(0)$ 和 $f(2)$ 都必须为0.

$$w_o(x, y) = \frac{w(x, y) - w(-x, -y)}{2}$$

若奇函数, 则 $f(0)$ 一定为0.

- 例3: $f=\{0,-1,0,1,0\}$
 - 非奇非偶

- 例4: 二维函数

奇函数

0	0	0	<u>0</u>	0	0
0	0	0	0	0	0
0	0	-1	0	1	0
0	0	-2	0	2	0
0	0	-1	0	1	0
0	0	0	0	0	0

二维DFT及其反变换的某些对称性质

TABLE 4.1

Some symmetry properties of the 2-D DFT and its inverse. $R(u,v)$ and $I(u,v)$ are the real and imaginary parts of $F(u,v)$, respectively. Use of the word *complex* indicates that a function has nonzero real and imaginary parts.

	Spatial Domain [†]	Frequency Domain [†]
1)	$f(x,y)$ real	$\Leftrightarrow F^*(u,v) = F(-u,-v)$
2)	$f(x,y)$ imaginary	$\Leftrightarrow F^*(-u,-v) = -F(u,v)$
3)	$f(x,y)$ real	$\Leftrightarrow R(u,v)$ even; $I(u,v)$ odd
4)	$f(x,y)$ imaginary	$\Leftrightarrow R(u,v)$ odd; $I(u,v)$ even
5)	$f(-x,-y)$ real	$\Leftrightarrow F^*(u,v)$ complex
6)	$f(-x,-y)$ complex	$\Leftrightarrow F(-u,-v)$ complex
7)	$f^*(x,y)$ complex	$\Leftrightarrow F^*(-u,-v)$ complex
8)	$f(x,y)$ real and even	$\Leftrightarrow F(u,v)$ real and even
9)	$f(x,y)$ real and odd	$\Leftrightarrow F(u,v)$ imaginary and odd
10)	$f(x,y)$ imaginary and even	$\Leftrightarrow F(u,v)$ imaginary and even
11)	$f(x,y)$ imaginary and odd	$\Leftrightarrow F(u,v)$ real and odd
12)	$f(x,y)$ complex and even	$\Leftrightarrow F(u,v)$ complex and even
13)	$f(x,y)$ complex and odd	$\Leftrightarrow F(u,v)$ complex and odd

[†]Recall that x, y, u , and v are *discrete* (integer) variables, with x and u in the range $[0, M-1]$, and y and v in the range $[0, N-1]$. To say that a complex function is *even* means that its real and imaginary parts are even, and similarly for an *odd* complex function. As before, " $\Leftrightarrow$ " indicates a Fourier transform pair.

性质举例:

TABLE 4.2

1-D examples of some of the properties in Table 4.1.

Property	$f(x)$	$F(u)$
3	$\{1, 2, 3, 4\}$	$\Leftrightarrow \{(10+0j), (-2+2j), (-2+0j), (-2-2j)\}$
4	$\{1j, 2j, 3j, 4j\}$	$\Leftrightarrow \{(0+2.5j), (.5-.5j), (0-.5j), (-.5-.5j)\}$
8	$\{2, 1, 1, 1\}$	$\Leftrightarrow \{5, 1, 1, 1\}$
9	$\{0, -1, 0, 1\}$	$\Leftrightarrow \{(0+0j), (0+2j), (0+0j), (0-2j)\}$
10	$\{2j, 1j, 1j, 1j\}$	$\Leftrightarrow \{5j, j, j, j\}$
11	$\{0j, -1j, 0j, 1j\}$	$\Leftrightarrow \{0, -2, 0, 2\}$
12	$\{(4+4j), (3+2j), (0+2j), (3+2j)\}$	$\Leftrightarrow \{(10+10j), (4+2j), (-2+2j), (4+2j)\}$
13	$\{(0+0j), (1+1j), (0+0j), (-1-j)\}$	$\Leftrightarrow \{(0+0j), (2-2j), (0+0j), (-2+2j)\}$

实函数共轭对称性的证明

- 因为
$$F(u + mN, v + nN) = \frac{1}{N} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \exp[-j2\pi(ux + vy) / N]$$

有
$$F^*(u + mN, v + nN) = \frac{1}{N} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \exp[j2\pi(ux + vy) / N]$$

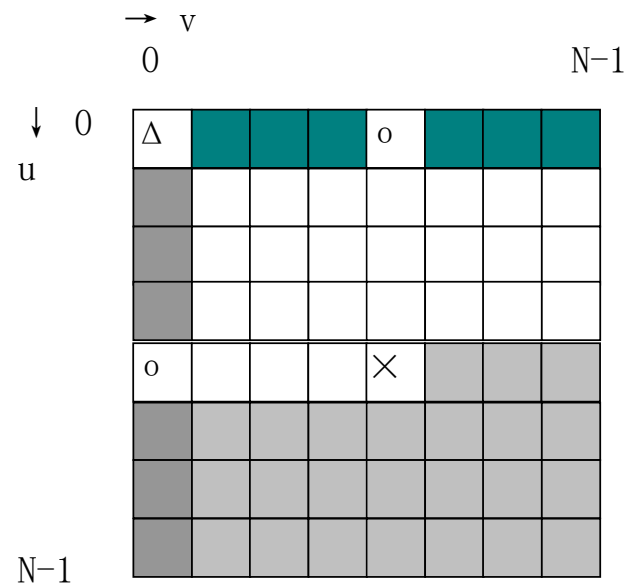
将其中u换成-u， v换成 -v， 有： $F^*(mN - u, nN - v) = F(u, v)$
于 $m, n=0, 1, 2, \dots$ 成立。

- 若 $m, n=1$ ， 则 $F^*(N - u, N - v) = F(u, v)$
对第0行， $u=0$ ： $F^*(N, N - v) = F(0, v)$
即： $F^*(0, N - v) = F(0, v)$

一般 $N=2^Q$ ， 故第0行关于 $v=N/2$ 点对称

对称性

- 同样对第**0**列也是关于 **$N/2$** 点对称
- 其它点关于 **$(N/2, N/2)$** 点对称



Fourier变换的共轭对称性

傅里叶谱和相角

- 二维**Fourier**变换的频谱为:

$$F(u, v) = |F(u, v)| \exp[j\phi(u, v)] = R(u, v) + jI(u, v)$$

- 幅度为: $|F(u, v)| = [R^2(u, v) + I^2(u, v)]^{1/2}$

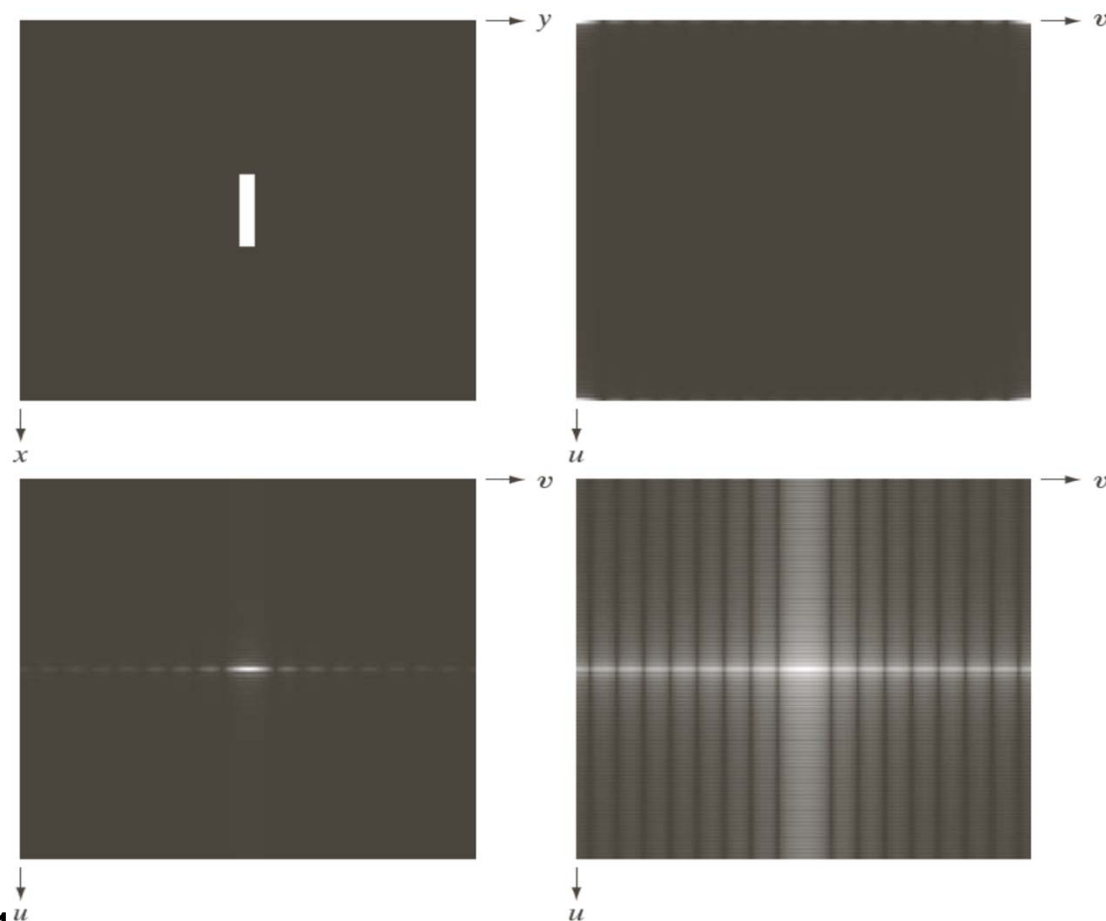
- 相位角为: $\phi(u, v) = \arctan\left[\frac{I(u, v)}{R(u, v)}\right]$

- 功率谱: $P(u, v) = |F(u, v)|^2 = R^2(u, v) + I^2(u, v)$

- 对于实函数的傅里叶变换: $|F(u, v)| = |F(-u, -v)|$

- $\phi(u, v) = -\phi(-u, -v)$

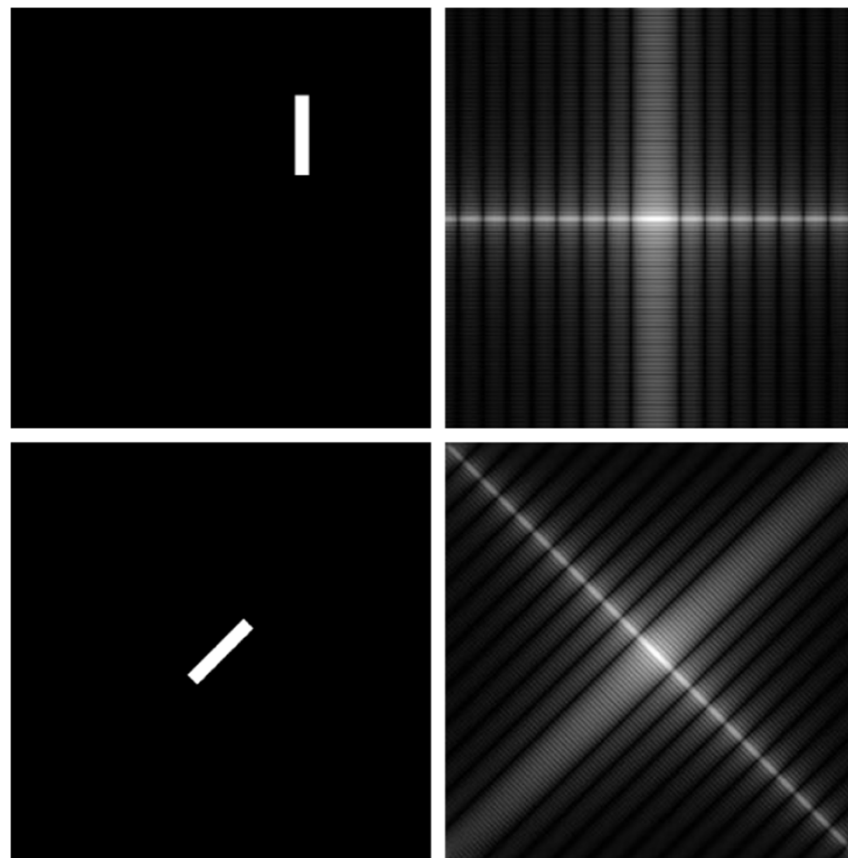
一个简单函数的傅里叶谱（幅度谱）



1原信号	2傅里叶谱
3中心移动后的傅里叶谱	4对数变换后的傅里叶谱 ($1+\log F $)

谱相对与原图平移和旋转的变化

- 1. 幅度谱对图像平移不敏感；
- 2. 幅度谱随图像的旋转而旋转。



图像平移或旋转后对应相角的变化

- 相位是各个正弦分量关于原点的位移的度量；
- 相位谱没有幅度谱那么直观，很少能得到直觉的信息；
- 相位携带了图像中可辨别物体的定位信息

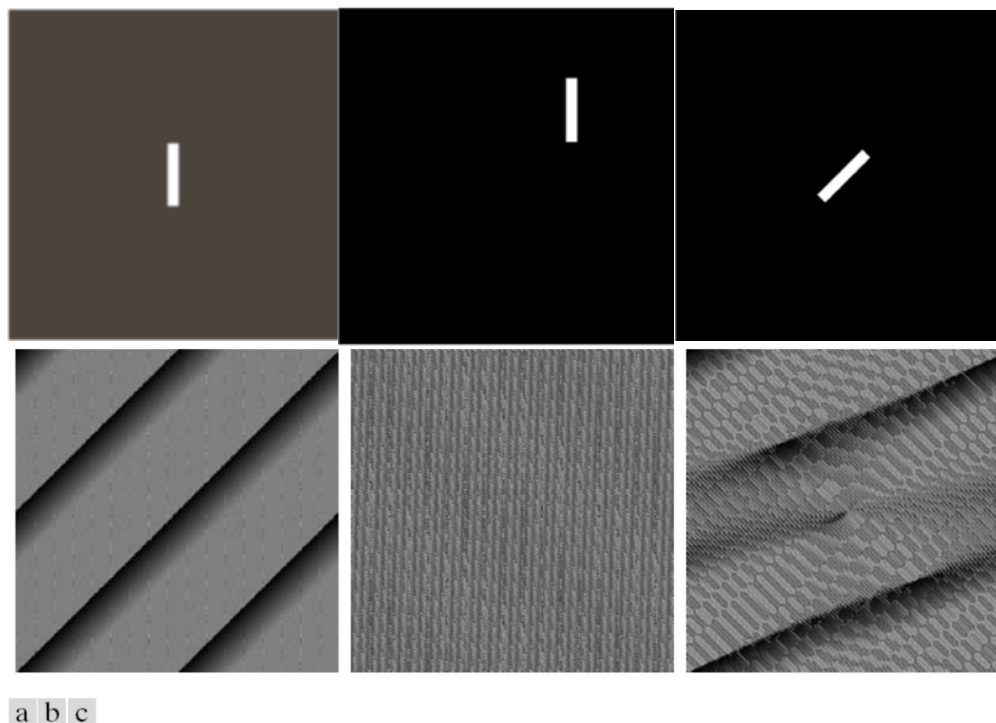
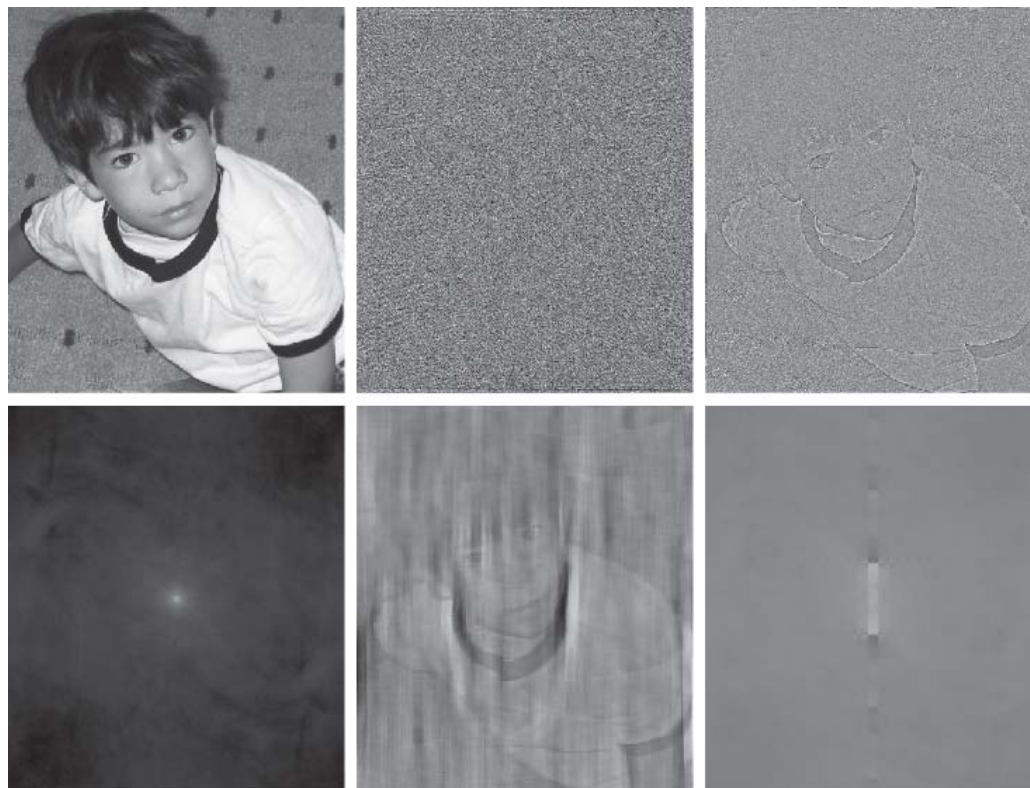


FIGURE 4.26 Phase angle array corresponding (a) to the image of the centered rectangle in Fig. 4.24(a), (b) to the translated image in Fig. 4.25(a), and (c) to the rotated image in Fig. 4.25(c).

相位的重要性



原图	相角	只用相角重构
只用谱重构	用自身相角和矩形白框的谱重建	用自身谱和矩形白框的相角重建

- 相比幅度谱，相位谱对图像内容起到更重要的支配性作用

- 7.尺度变换 (缩放)

- 对于常数a和b

$$af(x, y) \Leftrightarrow aF(u, v)$$

$$f(ax, by) \Leftrightarrow \frac{1}{|ab|} F\left(\frac{u}{a}, \frac{v}{b}\right)$$

- 8. 频谱的零频率分量

$$F(0,0) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) = N^2 \times \bar{f}$$

- 9 . 能量保持

$$\sum_{x=0}^{N-1} \sum_{y=0}^{N-1} |f(x, y)|^2 = \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} |F(u, v)|^2$$

二维卷积定理

- 二维卷积定义:

$$f(x, y) * h(x, y) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(x - m, y - n)$$

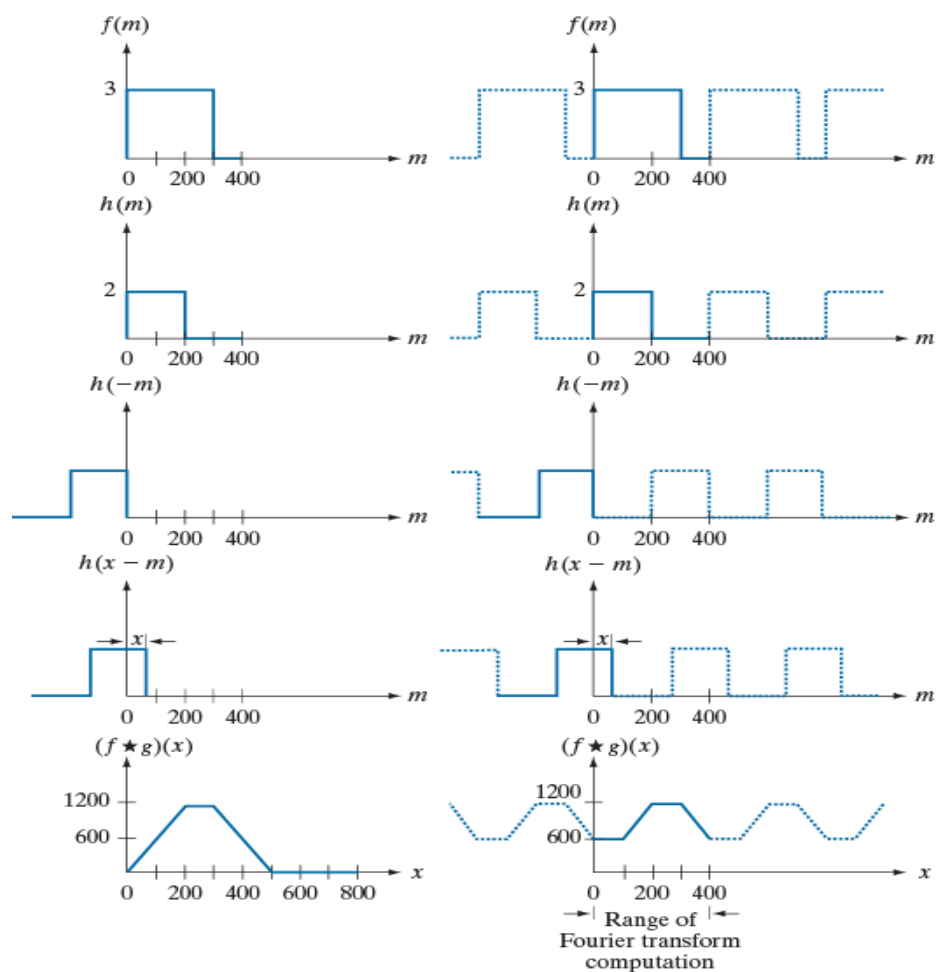
- 与傅里叶变换的关系

$$f(x, y) * h(x, y) = F^{-1}(F(u, v) H(u, v))$$

$$f(x, y) h(x, y) = F^{-1}(F(u, v) * H(u, v))$$

周期扩充（一维情况）

- 若要使DFT方法与直接卷积法产生相同的结果，则在DFT方法实施前，必须对函数补足周期后再计算变换，否则会产生缠绕错误。



零填充方法

- 设 $f(x,y)$ 和 $h(x,y)$ 的大小分别为 M_1*N_1 和 M_2*N_2 ，则
 $f(x,y)*h(x,y)$ 图像大小为 $M*N$ ， $M= M_1+M_2-1$ ， $N= N_1+N_2-1$
 - 将 $f(x,y)$ 和 $h(x,y)$ 补零扩展，使均为 $M*N$ 的图像，并且循环重复，记为 $f_e(x,y)$ 和 $g_e(x,y)$ ，之后采用卷积定理；
- 若原函数尾端不是零，补充0之后会有频谱泄漏，导致图像产生块效应。
 - 补救方法:采用高斯窗函数将降低泄漏。

傅里叶变换可分性

TABLE 4.3
Summary of DFT
definitions and
corresponding
expressions.

Name	Expression(s)
1) Discrete Fourier transform (DFT) of $f(x, y)$	$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$
2) Inverse discrete Fourier transform (IDFT) of $F(u, v)$	$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$
3) Spectrum	$ F(u, v) = [R^2(u, v) + I^2(u, v)]^{1/2} \quad R = \text{Real}(F); I = \text{Imag}(F)$
4) Phase angle	$\phi(u, v) = \tan^{-1} \left[\frac{I(u, v)}{R(u, v)} \right]$
5) Polar representation	$F(u, v) = F(u, v) e^{j\phi(u, v)}$
6) Power spectrum	$P(u, v) = F(u, v) ^2$
7) Average value	$\bar{f} = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) = \frac{1}{MN} F(0, 0)$
8) Periodicity (k_1 and k_2 are integers)	$F(u, v) = F(u + k_1 M, v) = F(u, v + k_2 N)$ $= F(u + k_1, v + k_2 N)$ $f(x, y) = f(x + k_1 M, y) = f(x, y + k_2 N)$ $= f(x + k_1, y + k_2 N)$
9) Convolution	$(f \star h)(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(x - m, y - n)$
10) Correlation	$(f \star h)(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f^*(m, n) h(x + m, y + n)$
11) Separability	The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.
12) Obtaining the IDFT using a DFT algorithm	$MNf^*(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v) e^{-j2\pi(ux/M + vy/N)}$ This equation indicates that inputting $F^*(u, v)$ into an algorithm that computes the forward transform (right side of above equation) yields $MNf^*(x, y)$. Taking the complex conjugate and dividing by MN gives the desired inverse. See Section 4.11.

2D傅里叶变换的可分性



利用正变换获得反变换



二维傅里叶变换对性质小结1

Name	DFT Pairs
1) Symmetry properties	See Table 4.1
2) Linearity	$af_1(x, y) + bf_2(x, y) \Leftrightarrow aF_1(u, v) + bF_2(u, v)$
3) Translation (general)	$f(x, y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u - u_0, v - v_0)$ $f(x - x_0, y - y_0) \Leftrightarrow F(u, v)e^{-j2\pi(ux_0/M+vy_0/N)}$
4) Translation to center of the frequency rectangle, $(M/2, N/2)$	$f(x, y)(-1)^{x+y} \Leftrightarrow F(u - M/2, v - N/2)$ $f(x - M/2, y - N/2) \Leftrightarrow F(u, v)(-1)^{u+v}$
5) Rotation	$f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \varphi + \theta_0)$ $x = r \cos \theta \quad y = r \sin \theta \quad u = \omega \cos \varphi \quad v = \omega \sin \varphi$
6) Convolution theorem [†]	$f(x, y) \star h(x, y) \Leftrightarrow F(u, v)H(u, v)$ $f(x, y)h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$

← 线性变换

← 不是移不变变换！！

(Continued)

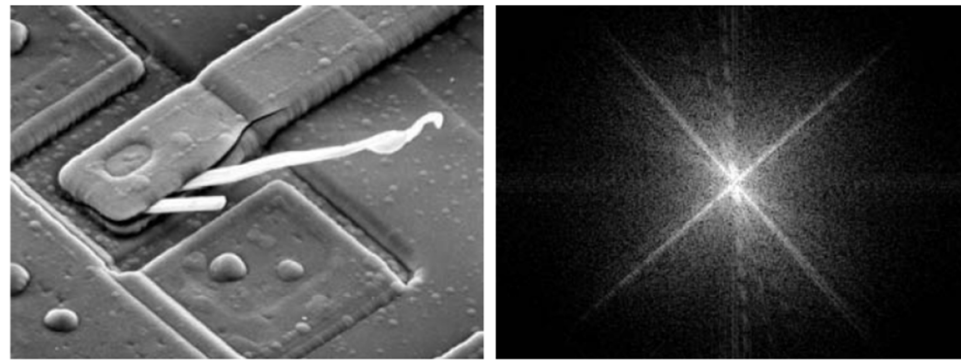
二维傅里叶变换对性质小结2

Name	DFT Pairs
7) Correlation theorem [†]	$f(x, y) \star h(x, y) \Leftrightarrow F^*(u, v) H(u, v)$ $f^*(x, y) h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$
8) Discrete unit impulse	$\delta(x, y) \Leftrightarrow 1$
9) Rectangle	$\text{rect}[a, b] \Leftrightarrow ab \frac{\sin(\pi ua)}{(\pi ua)} \frac{\sin(\pi vb)}{(\pi vb)} e^{-j\pi(ua+vb)}$
10) Sine	$\sin(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$ $j \frac{1}{2} [\delta(u + Mu_0, v + Nv_0) - \delta(u - Mu_0, v - Nv_0)]$
11) Cosine	$\cos(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$ $\frac{1}{2} [\delta(u + Mu_0, v + Nv_0) + \delta(u - Mu_0, v - Nv_0)]$
The following Fourier transform pairs are derivable only for continuous variables, denoted as before by t and z for spatial variables and by μ and ν for frequency variables. These results can be used for DFT work by sampling the continuous forms.	
12) Differentiation (The expressions on the right assume that $f(\pm\infty, \pm\infty) = 0$.)	$\left(\frac{\partial}{\partial t}\right)^m \left(\frac{\partial}{\partial z}\right)^n f(t, z) \Leftrightarrow (j2\pi\mu)^m (j2\pi\nu)^n F(\mu, \nu)$ $\frac{\partial^m f(t, z)}{\partial t^m} \Leftrightarrow (j2\pi\mu)^m F(\mu, \nu); \frac{\partial^n f(t, z)}{\partial z^n} \Leftrightarrow (j2\pi\nu)^n F(\mu, \nu)$
13) Gaussian	$A 2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2+z^2)} \Leftrightarrow A e^{-(\mu^2+\nu^2)/2\sigma^2}$ (A is a constant)

[†] Assumes that the functions have been extended by zero padding. Convolution and correlation are associative, commutative, and distributive.

频率域滤波基础

- 想将图像的某些特定分量与其变换直接联系起来是不可能的；
- 但可以在一般意义上，做出一些联系。比如将傅里叶变换的频率和图像中的亮度变化模式联系起来。
 - 图像中的边缘方向、白色条带等在频谱图分布方向上的反映：
频谱图中正负45度方向的谱对应图像中正负45度方向的边缘；中心垂直虚线状谱对应凸起白条，且周期与白条宽度有关。



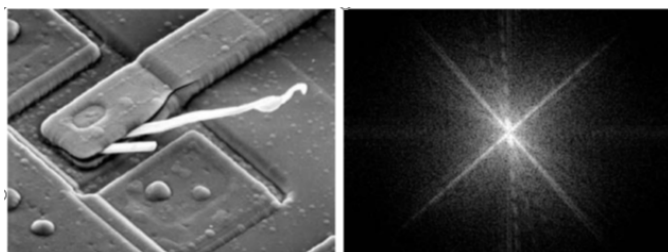
a b

FIGURE 4.29 (a) SEM image of a damaged integrated circuit. (b) Fourier spectrum of (a). (Original image courtesy of Dr. J. M. Hudak, Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, Canada.)

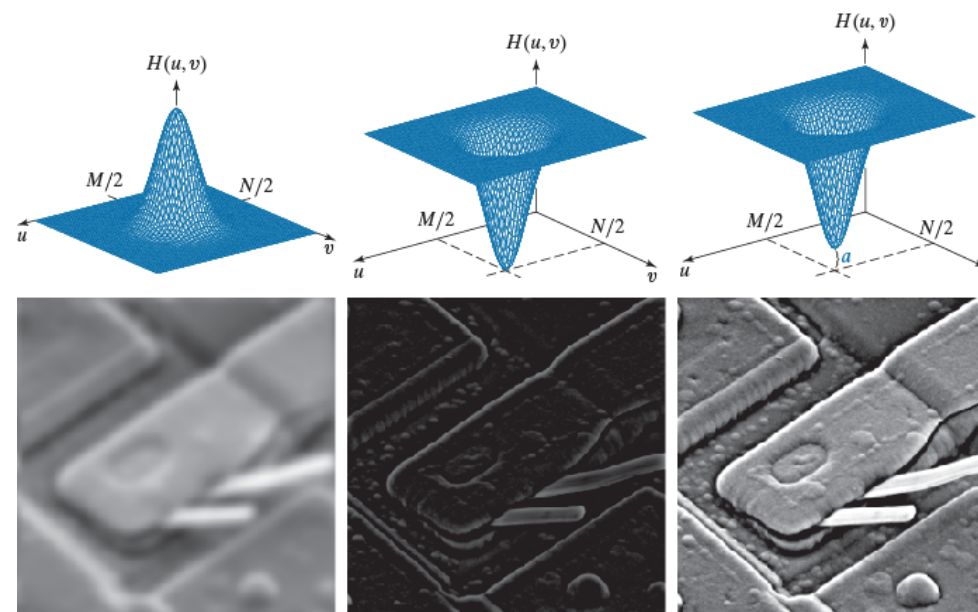
频率域滤波的原理

滤波原理: $g(x, y) = F^{-1}[H(u, v)F(u, v)]$

不同形状频率域滤波器的对应效果:



原始图像与频谱



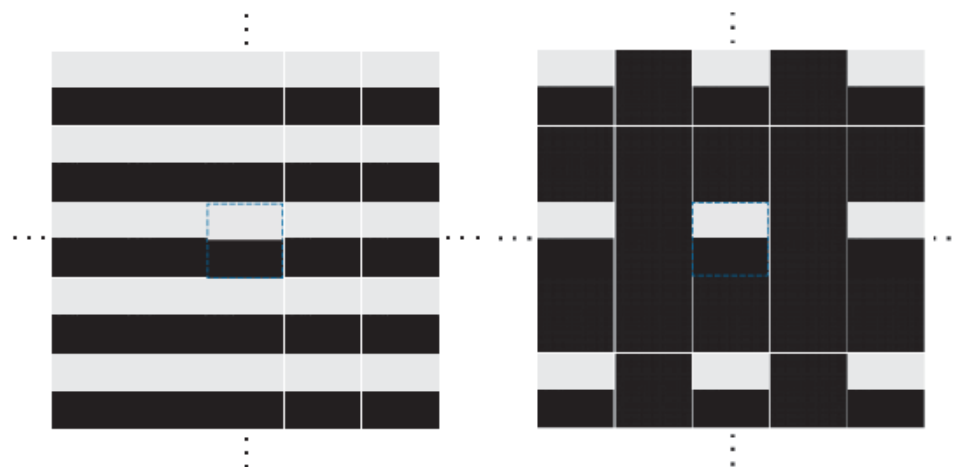
频率域滤波前输入图像有无 (空域) 零填充的效果对比



a b c

FIGURE 4.31 (a) A simple image. (b) Result of blurring with a Gaussian lowpass filter without padding. (c) Result of lowpass filtering with zero padding. Compare the vertical edges in (b) and (c).

- 左上：对比
- 右下：原始周期函数
和对应的零填充后的函数

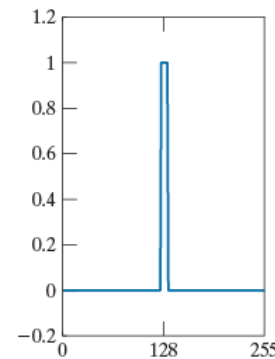


频率域滤波器的（空域）零填充

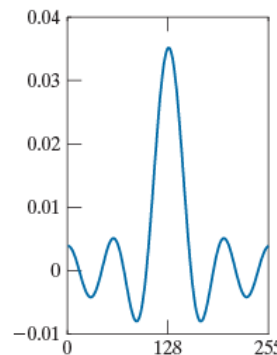
（使得模板大小与填零后的空域信号一样大）

- 候选方法：
 - 频率域构建与图像尺寸相同的滤波器，然后IDFT得到空间滤波器，在空域零填充，再DFT返回频率域。
- 按照候选方法获得填零后的空间滤波器，在频域中存在振铃现象

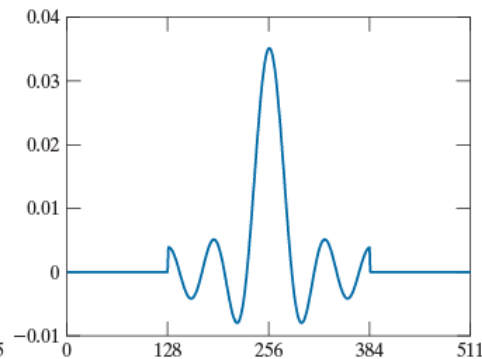
1. 频域滤波理想滤波器



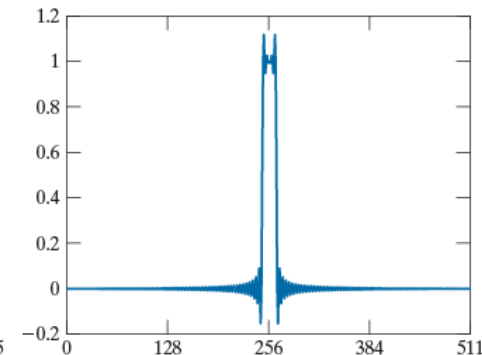
2. 对应空域滤波器



3. 填零后空域滤波器



4. 填零后频域滤波器



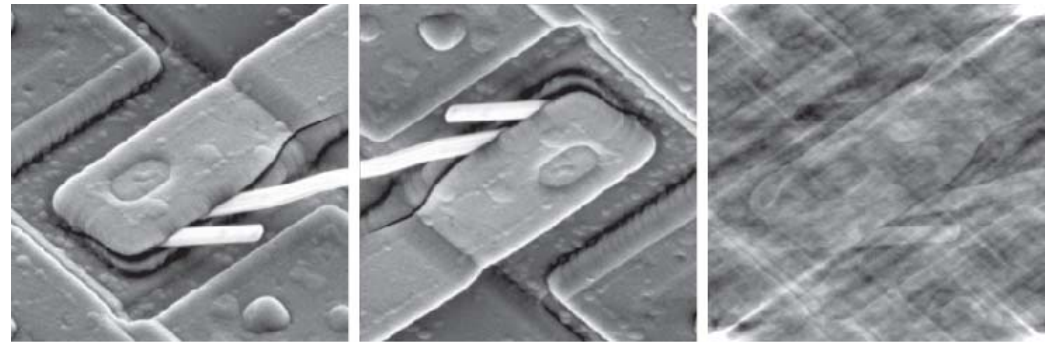
频率域滤波的实际处理方法

- 实际中为减缓振铃现象，可采取的频率域滤波方法：
 - 把图像填充为 $P*Q$ 大小(若原图和原始滤波器大小均为 $N*N$ ，则 $P=Q=2N$)，直接构造 $P*Q$ 大小的频域滤波器（此处的频域滤波器没有截断，没有振铃），然后相乘（注意频率域滤波必须尺寸一样）。
 - 虽然仍然存在卷绕误差（滤波器本身未零填充造成），但错误可以通过图像填充提供的零间隔有效减轻，振铃效应也更轻。

零相移滤波器

$$\begin{aligned} g(x, y) &= F^{-1}[H(u, v)F(u, v)] \\ &= F^{-1}[H(u, v)R(u, v) + jH(u, v)I(u, v)] \end{aligned}$$

- 等同地影响实部和虚部而不影响相位的滤波器称为零相移滤波器。
- 即使相位角有很小的变化，也会对滤波输出有很大的影响。



- 幅度谱不变，对相位角分别乘以-1和0.25后恢复的图像。

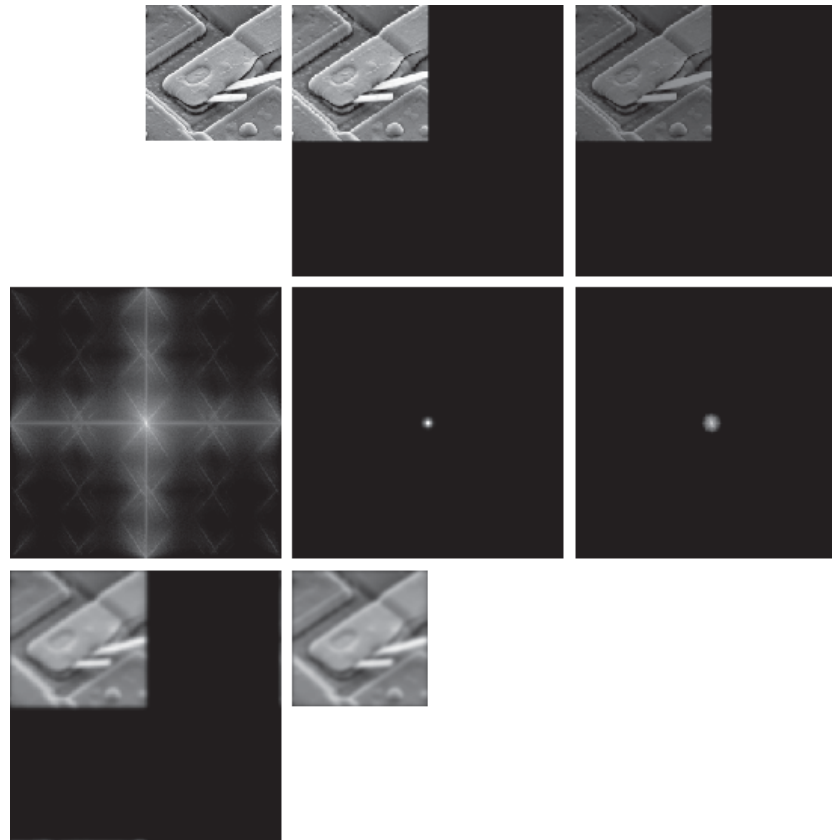
频率域滤波步骤小结

- 1. 将大小为 $M*N$ 的图像 $f(x,y)$, 根据填零规则扩展为 $P*Q$, 典型地, $P=2*M$, $Q=2*N$, 得到填零后的图像 $f_p(x,y)$;
- 2. 用 $(-1)^{x+y} f_p(x,y)$ 将图像平移;
- 3. 计算来自上一步函数的DFT, 得到 $F(u,v)$;
- 4. 生成一个实对称的, 大小也为 $P*Q$ 的, 中心在 $(P/2, Q/2)$ 处的滤波函数 $H(u,v)$, 并与 $F(u,v)$ 相乘;
- 5. 得到处理后的图像

$$g_p(x,y) = \{real[F^{-1}(H(u,v)F(u,v))]\}(-1)^{x+y}$$

- 6. 提取 $g_p(x,y)$ 的左上限 $M*N$ 区域, 即为滤波后的结果

频率域滤波步骤实例



a b c
d e f
g h

FIGURE 4.35

(a) An $M \times N$ image, f .
(b) Padded image, f_p , of size $P \times Q$.
(c) Result of multiplying f_p by $(-1)^{x+y}$.
(d) Spectrum of F . (e) Centered Gaussian lowpass filter transfer function, H , of size $P \times Q$.
(f) Spectrum of the product HF .
(g) Image g_p , the real part of the IDFT of HF , multiplied by $(-1)^{x+y}$.
(h) Final result, g , obtained by extracting the first M rows and N columns of g_p .

空间和频率域滤波之间的对应

- 空间域滤波器和频率域滤波器是一对傅里叶变换对；

$$h(x, y) \Leftrightarrow H(u, v)$$

- 由于空间域滤波器（空域）也对应于频率域滤波器对单位冲激的响应（频域），因此 $h(x, y)$ 也称为 $H(u, v)$ 的冲激响应；
- 实际当中 $h(x, y)$ 的数值个数有限，因此称为有限冲激响应（FIR）滤波器，是线性的。
- 在DFT中，滤波实际上是相同尺寸函数的循环卷积；而在空域中，滤波是不同尺寸函数的普通卷积。
- DFT是线性变换，因此只能实现线性滤波；而空域中可以实现非线性滤波。

空间和频率域滤波之间的对应

- 例：一维高斯滤波器

$$H(u) = Ae^{-u^2/2\sigma^2}$$

$$h(x) = \sqrt{2\pi}\sigma Ae^{-2\pi^2\sigma^2x^2}$$

- 特点：

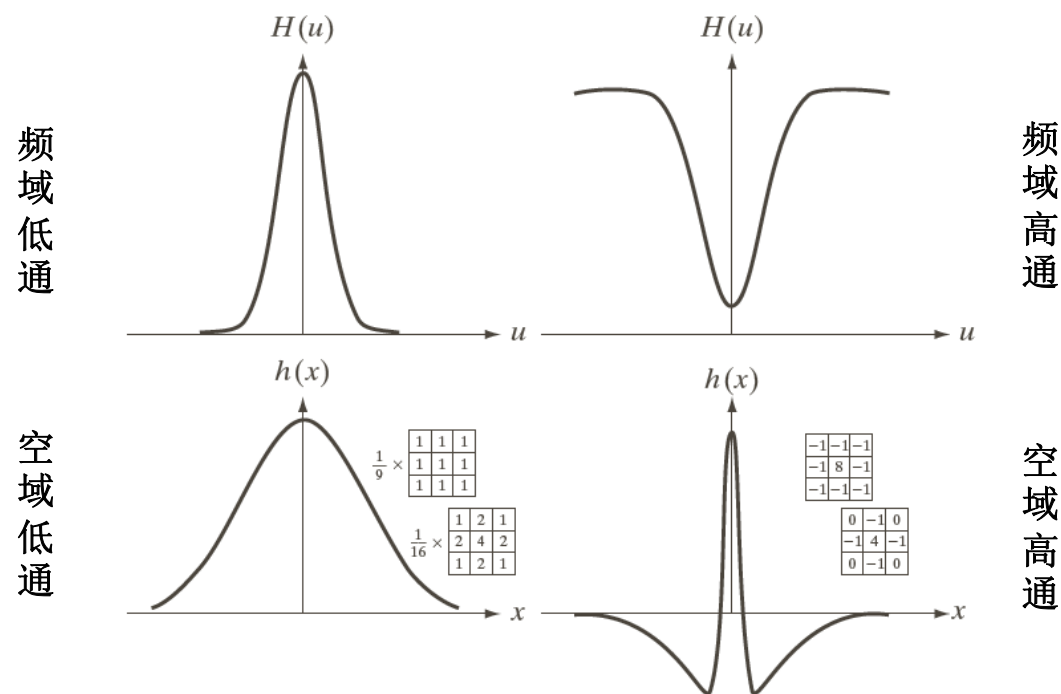
- （1）空域和频域都是高斯、实函数，直观易于操作；
- （2）函数表现为互易的，从参数 σ 分别在分母和分子中可以看出。当 $H(u)$ 有较宽的外形时， $h(x)$ 有较窄的外形。

由高斯差构造一个高通（带通）滤波器

- 常见做法：可以由频率域先构造频率域滤波器，然后返回空域用小模板进行空域滤波。

$$H(u) = Ae^{-u^2/2\sigma_1^2} - Be^{-u^2/2\sigma_2^2}, \quad s.t. A \geq B, \quad \sigma_1 > \sigma_2$$

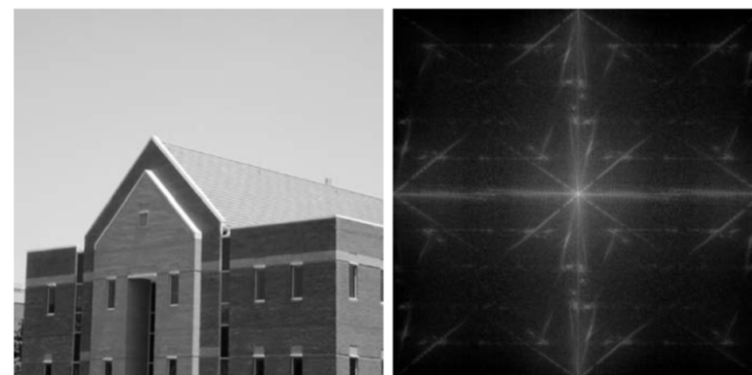
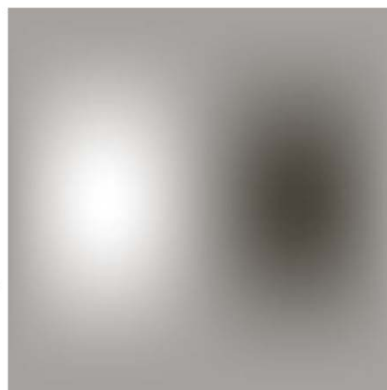
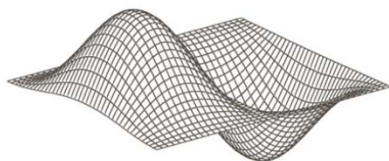
$$h(x) = \sqrt{2\pi}A\sigma_1 e^{-2\pi^2\sigma_1^2 x^2} - \sqrt{2\pi}B\sigma_2 e^{-2\pi^2\sigma_2^2 x^2}$$



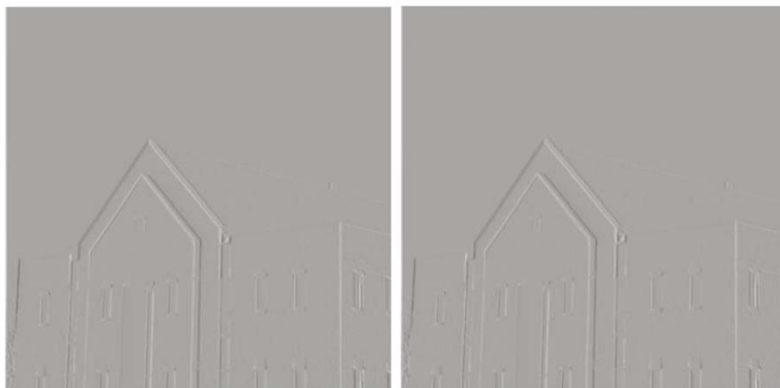
由小空间模板得到频率域滤波器

- 例：Sobel模板

-1	0	1
-2	0	2
-1	0	1



输入图像	频谱
------	----



Sobel模板	滤波器频谱
频域滤波结果	空域滤波结果

图像的频域平滑

- 滤除白噪声

频域滤波表示: $G(u, v) = H(u, v)F(u, v)$

$H(u, v)$ 是滤波器的响应,或称转移函数、传递函数

低通滤波能否提高信噪比,为什么?

低通滤波是否使图像模糊,为什么?

1. 理想低通滤波器

$$H(u,v) = \begin{cases} 1 & , D(u,v) \leq D_0 \\ 0 & , D(u,v) > D_0 \end{cases}$$

其中 $D(u,v) = (u^2 + v^2)^{\frac{1}{2}}$ 是频域中点到原点的距离

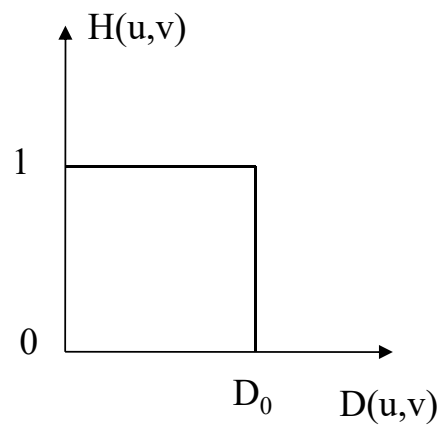
D_0 是一非负实数,称为截止频率

缺点:

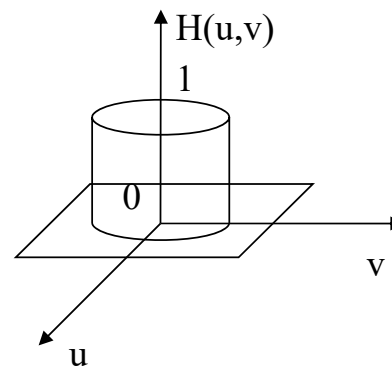
(1) 边沿模糊

(2) 振铃现象

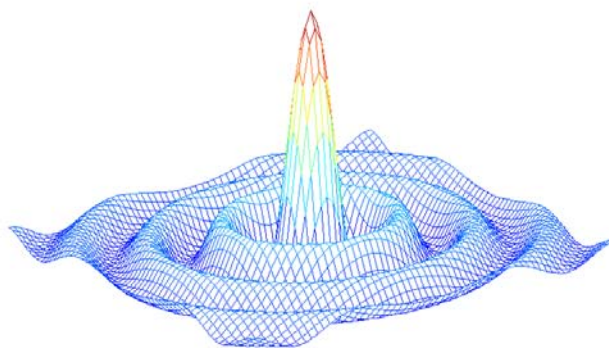
理想低通滤波器



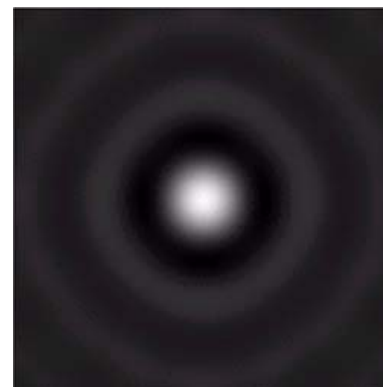
(a)理想低通的截面图



(b)透视图

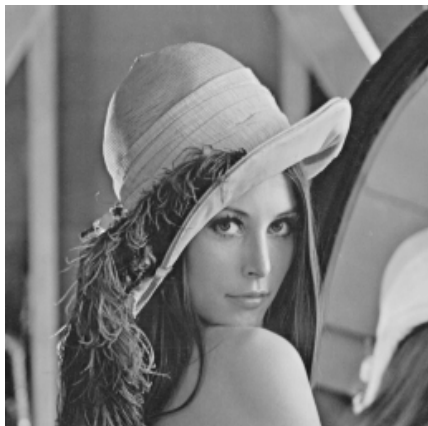


(c) 理想低通的冲激响应

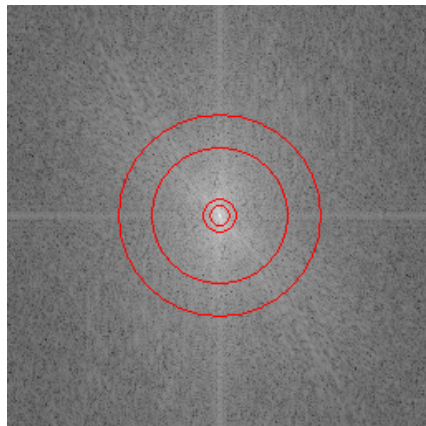


(d) 冲激响应的图像显示

理想低通滤波器



(a)Lenna原图



(b)频谱图



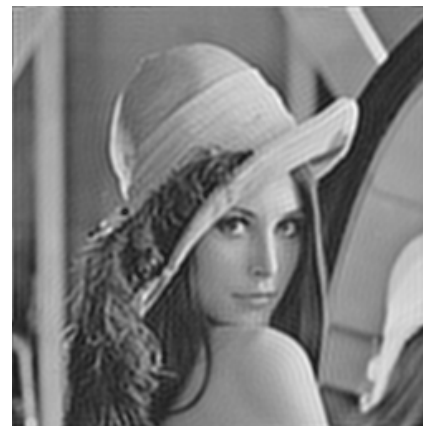
(c)90%能量



(d)95%能量

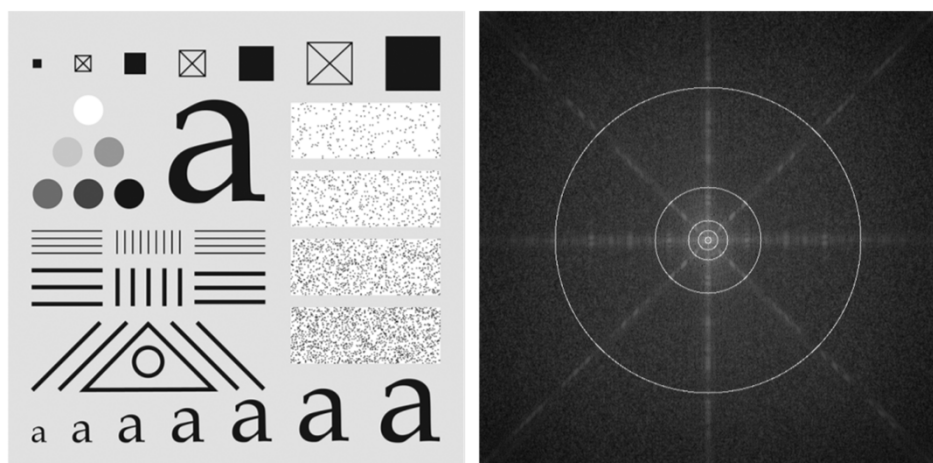


(e)99%能量（依然有振铃，
肩膀部位）
理想低通滤波效果



(d)99.5%能量

理想低通平滑示例



a b

FIGURE 4.40 (a) Test pattern of size 688×688 pixels, and (b) its spectrum. The spectrum is double the image size as a result of padding, but is shown half size to fit. The circles have radii of 10, 30, 60, 160, and 460 pixels with respect to the full-size spectrum. The radii enclose 86.9, 92.8, 95.1, 97.6, and 99.4% of the padded image power, respectively.



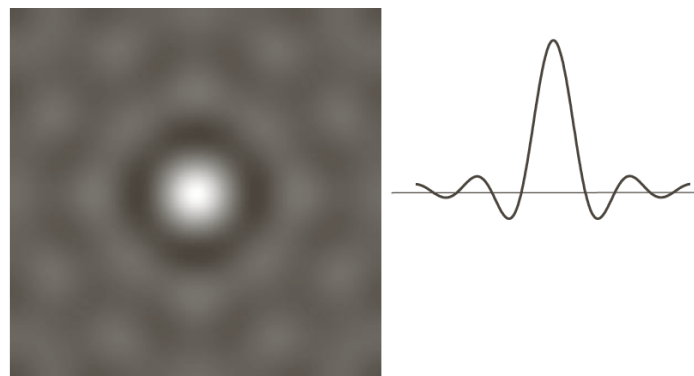
a b c
d e f

FIGURE 4.41 (a) Original image of size 688×688 pixels. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.40(b). The power removed by these filters was 13.1, 7.2, 4.9, 2.4, and 0.6% of the total, respectively. We used mirror padding to avoid the black borders characteristic of zero padding, as illustrated in Fig. 4.31(c).

振铃！！

理想低通的特性

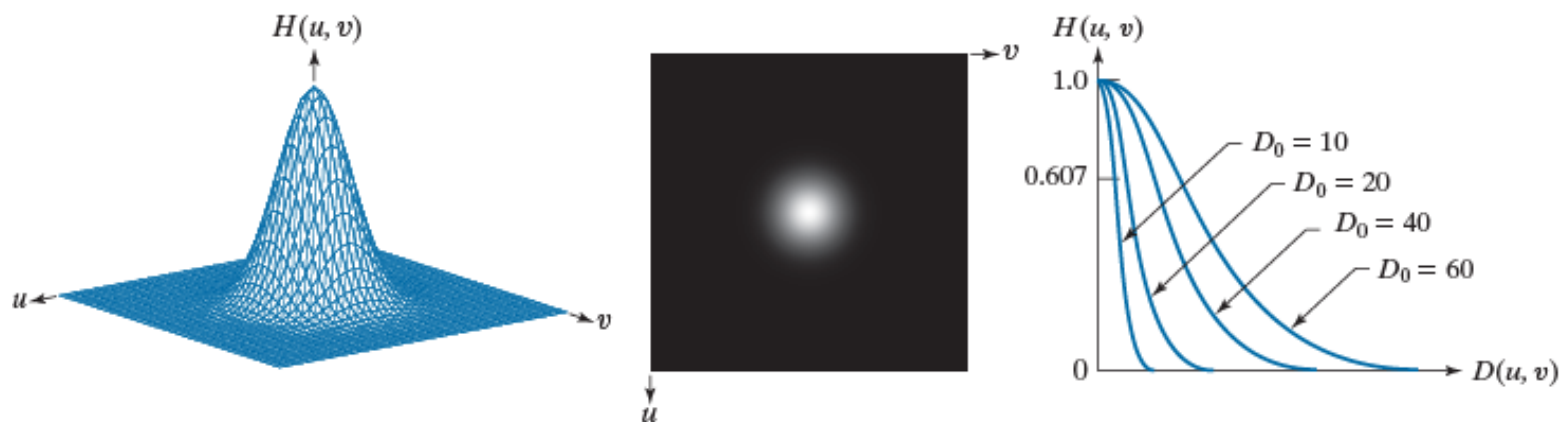
- 在二维空域中的类似sinc波形中，中心主瓣是引起模糊的主要原因，旁瓣是引起振铃的主要原因。
- sinc函数的展开度与滤波截止频率的大小成反比。



高斯低通滤波器

$$H(u, v) = e^{-\frac{D(u, v)^2}{2\sigma^2}} = e^{-\frac{D(u, v)^2}{2D_0^2}}$$

- 当 $D(u, v) = D_0$ 时，高斯低通下降到最大值的**0.607**处。
- 频率域高斯的傅里叶变换也是高斯函数，因此没有振铃效应，可以应用到医学等特殊场合。

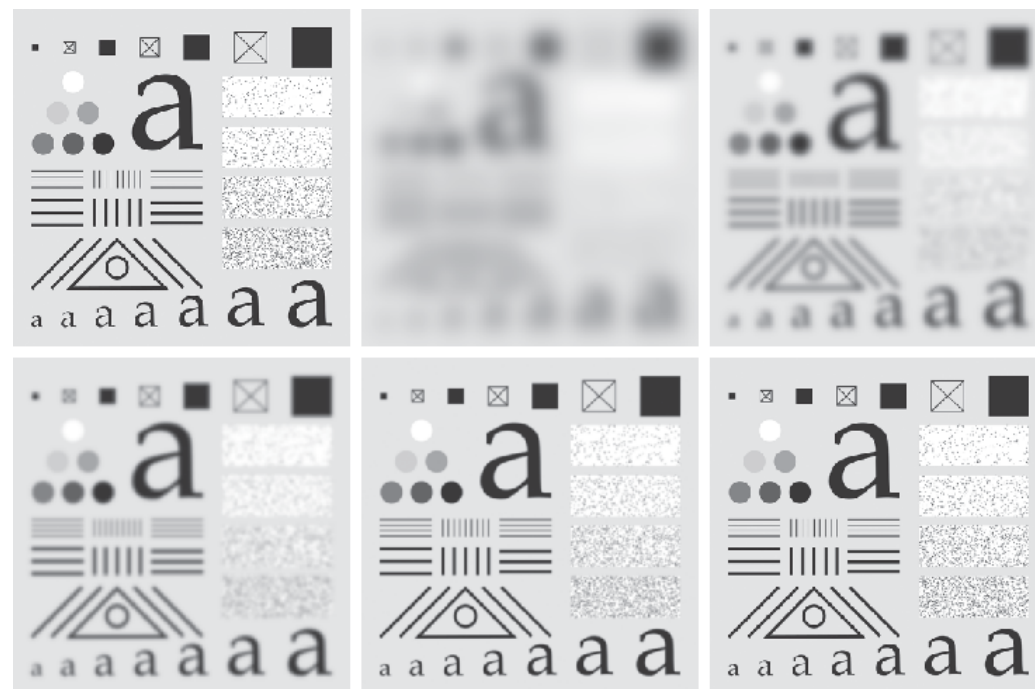


高斯低通效果



a b c

FIGURE 4.50 (a) Original image (784×732 pixels). (b) Result of filtering using a GLPF with $D_0 = 100$. (c) Result of filtering using a GLPF with $D_0 = 80$. Note the reduction in fine skin lines in the magnified sections in (b) and (c).



a b c
d e f

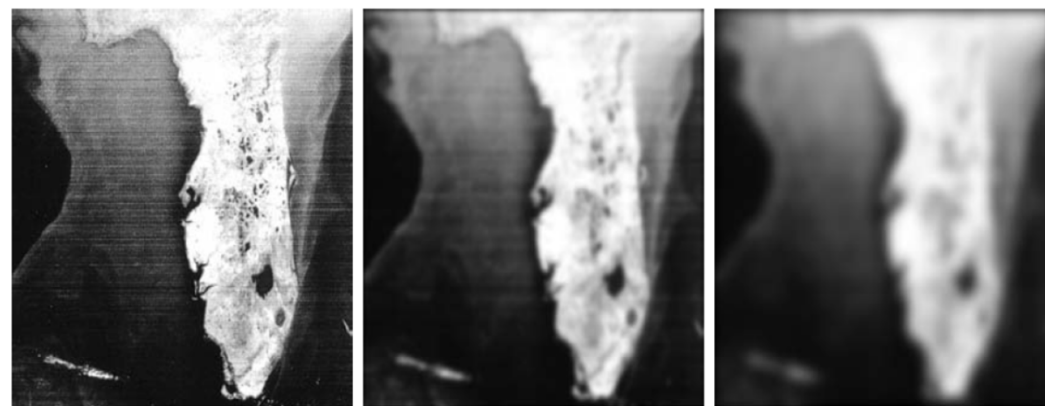
FIGURE 4.44 (a) Original image of size 688×688 pixels. (b)–(f) Results of filtering using GLPFs with cutoff frequencies at the radii shown in Fig. 4.40. Compare with Fig. 4.41. We used mirror padding to avoid the black borders characteristic of zero padding.

高斯低通应用示例

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



a b c

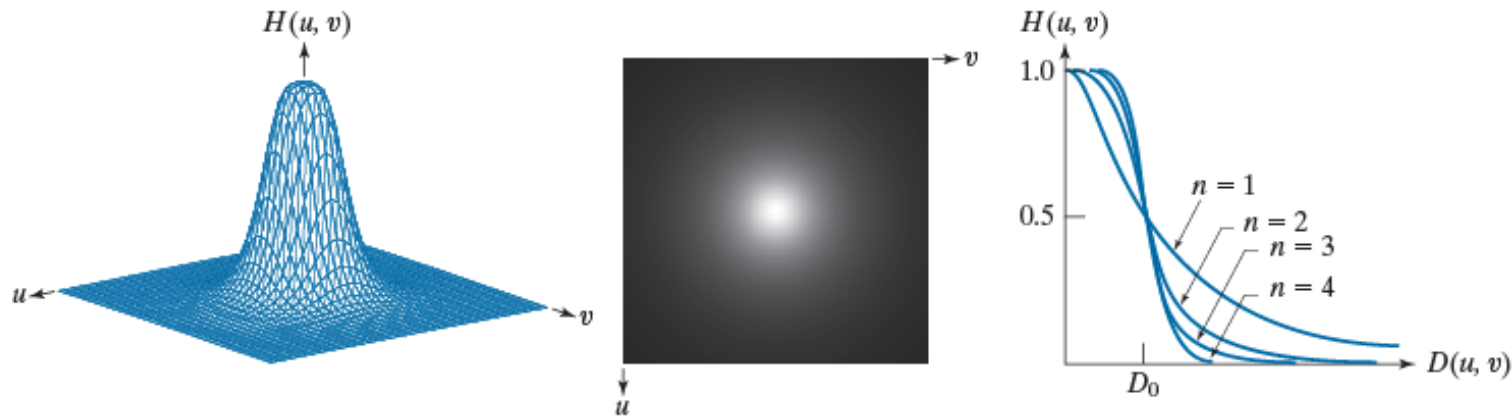
FIGURE 4.51 (a) Image showing prominent horizontal scan lines. (b) Result of filtering using a GLPF with $D_0 = 50$. (c) Result of using a GLPF with $D_0 = 20$. (Original image courtesy of NOAA.)

巴特沃斯(Butterworth)低通滤波器

- **n**阶巴特沃斯低通滤波器

$$H(u, v) = \frac{1}{1 + \left[\frac{D(u, v)}{D_0} \right]^{2n}}$$

- 频率特性较平滑，振铃现象较少,但随**n**增加,振铃会加重



巴特沃斯低通平滑

- 示例



(a) 原图被均方差为20的Gauss噪声污染

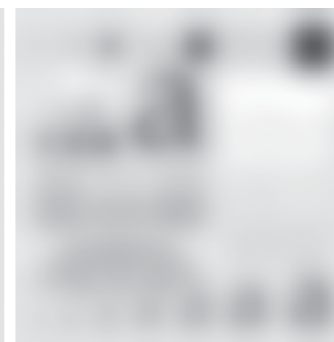


(b) 理想低通滤波



(c) Butterworth低通滤波(n=1)

对高斯噪声污染的图像的低通滤波



巴特沃斯低通振铃与阶数的关系

- 阶数越高，振铃越严重；
- 2阶是较好的折中。

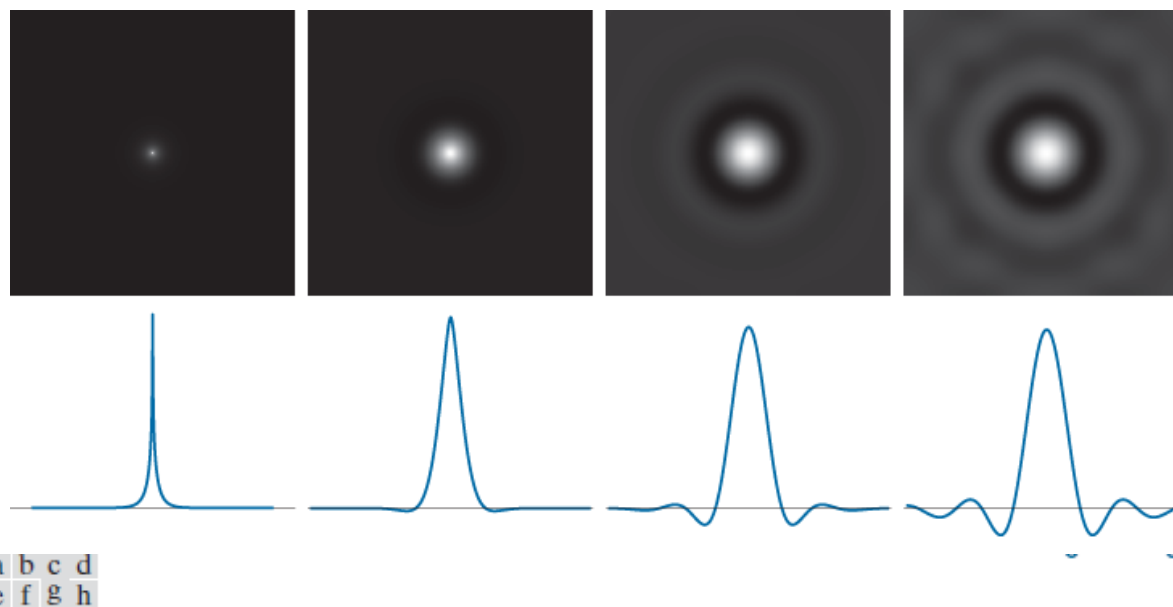


FIGURE 4.47 (a)–(d) Spatial representations (i.e., spatial kernels) corresponding to BLPF transfer functions of 1000×1000 pixels, cut-off frequency of 5, and order 1, 2, 5, and 20, respectively. (e)–(h) Corresponding intensity profiles through the center of the filter functions.

三种低通的对比

TABLE 4.5

Lowpass filter transfer functions. D_0 is the cutoff frequency, and n is the order of the Butterworth filter.

Ideal	Gaussian	Butterworth
$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \leq D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$	$H(u,v) = e^{-D^2(u,v)/2D_0^2}$	$H(u,v) = \frac{1}{1 + [D(u,v)/D_0]^{2n}}$

- 巴特沃斯低通可以看做是介于理想低通和高斯低通之间的滤波器，当阶数较高时更趋近与理想低通，较低时趋近与高斯低通。

使用频率域滤波器锐化图像

频域滤波: $G(u, v) = H(u, v)F(u, v)$

与低通的关系 $H_{hp}(u, v) = 1 - H_{lp}(u, v)$

理想高通 $H(u, v) = \begin{cases} 0 & , \text{ 当 } D(u, v) \leq D_0 \\ 1 & , \text{ 当 } D(u, v) > D_0 \end{cases}$

巴特沃斯高通滤波 $H(u, v) = \frac{1}{1 + \left[\frac{D_0}{D(u, v)} \right]^{2n}}$

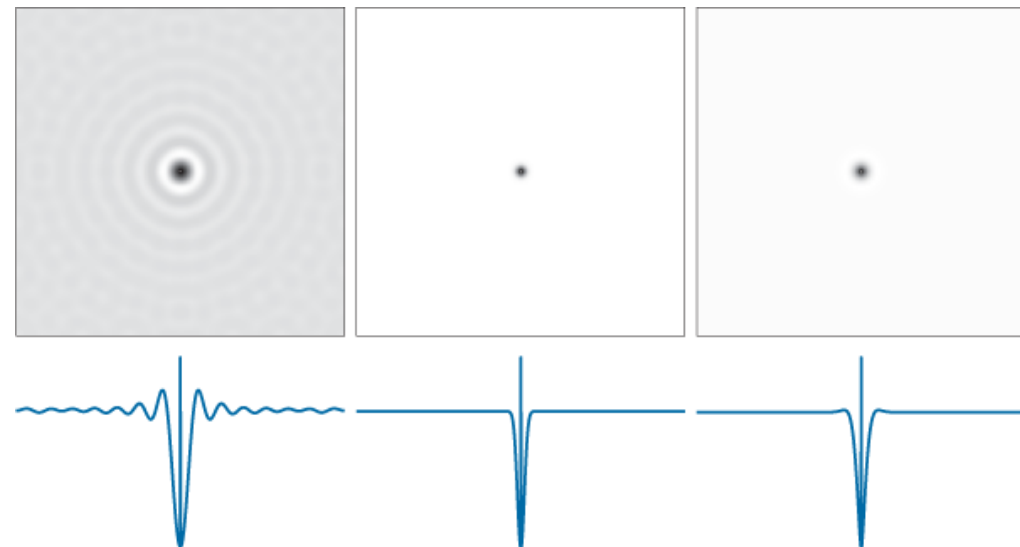
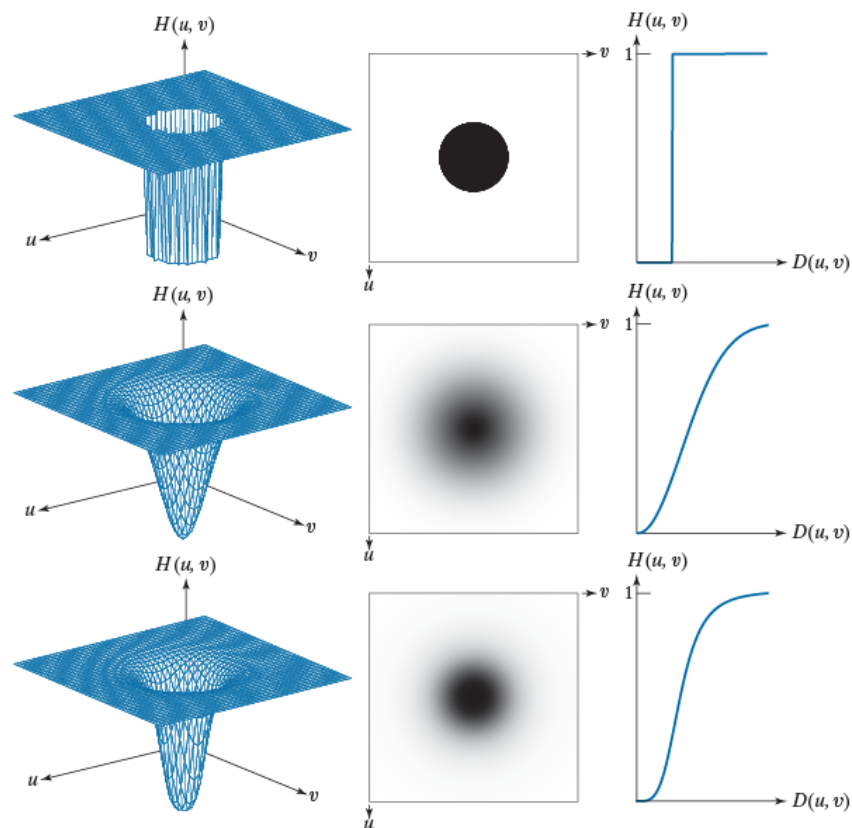
高斯高通滤波器 $H(u, v) = 1 - e^{-\frac{D^2(u, v)}{2D_0^2}}$

三种高通滤波器频域与空域表示

a b c
d e f
g h i

FIGURE 4.51

Top row: Perspective plot, image, and, radial cross section of an IHPF transfer function. Middle and bottom rows: The same sequence for GHPF and BHPF transfer functions. (The thin image borders were added for clarity. They are not part of the data.)



a b c
d e f

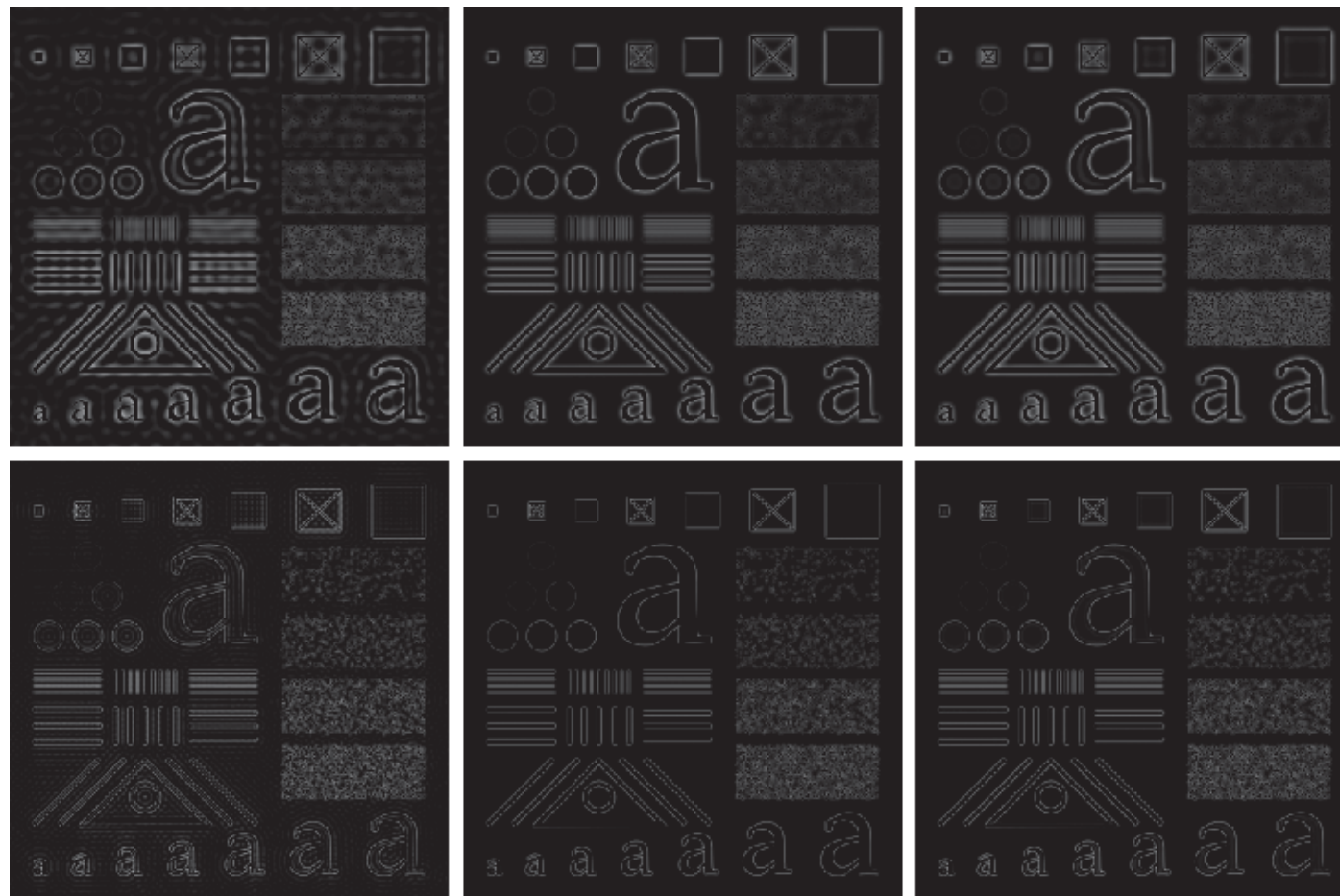
FIGURE 4.52 (a)–(c): Ideal, Gaussian, and Butterworth highpass spatial kernels obtained from IHPF, GHPF, and BHPF frequency-domain transfer functions. (The thin image borders are not part of the data.) (d)–(f): Horizontal intensity profiles through the centers of the kernels.

三种典型高通滤波器比较

TABLE 4.6

Highpass filter transfer functions. D_0 is the cutoff frequency and n is the order of the Butterworth transfer function.

Ideal	Gaussian	Butterworth
$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \leq D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$	$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$	$H(u,v) = \frac{1}{1 + [D_0/D(u,v)]^{2n}}$



a	b	c
d	e	f

FIGURE 4.53 Top row: The image from Fig. 4.40(a) filtered with IHPF, GHPF, and BHPF transfer functions using $D_0 = 60$ in all cases ($n = 2$ for the BHPF). Second row: Same sequence, but using $D_0 = 160$.

用高通滤波和阈值法增强图像

- 增强脊线，减少慢变污染



a b c

FIGURE 4.57 (a) Thumb print. (b) Result of highpass filtering (a). (c) Result of thresholding (b). (Original image courtesy of the U.S. National Institute of Standards and Technology.)

频率域的拉普拉斯算子

- 空域

$$\nabla^2 f = \frac{\partial^2 f}{\partial^2 x} + \frac{\partial^2 f}{\partial^2 y}$$

- 频率域

$$H(u, v) = -4\pi^2(u^2 + v^2) = -4\pi^2 D^2(u, v)$$

- 频率域增强:

$$\begin{aligned} g(x, y) &= F^{-1}\{F(u, v) - H(u, v)F(u, v)\} \\ &= F^{-1}\{[1 + 4\pi^2 D^2(u, v)]F(u, v)\} \end{aligned}$$

频率域拉普拉斯增强



a b

FIGURE 4.58

(a) Original, blurry image.

(b) Image enhanced using the Laplacian in the frequency domain. Compare with Fig. 3.38(e).

高提升与高频强调滤波

- 空域钝化模板与高提升滤波

$$g_{mask}(x,y) = f(x,y) - \bar{f}(x,y)$$

$$g(x,y) = f(x,y) + k * g_{mask}(x,y)$$

- 对应频域处理

$$f_{LP}(x,y) = F^{-1}[H_{LP}(u,v)F(u,v)]$$

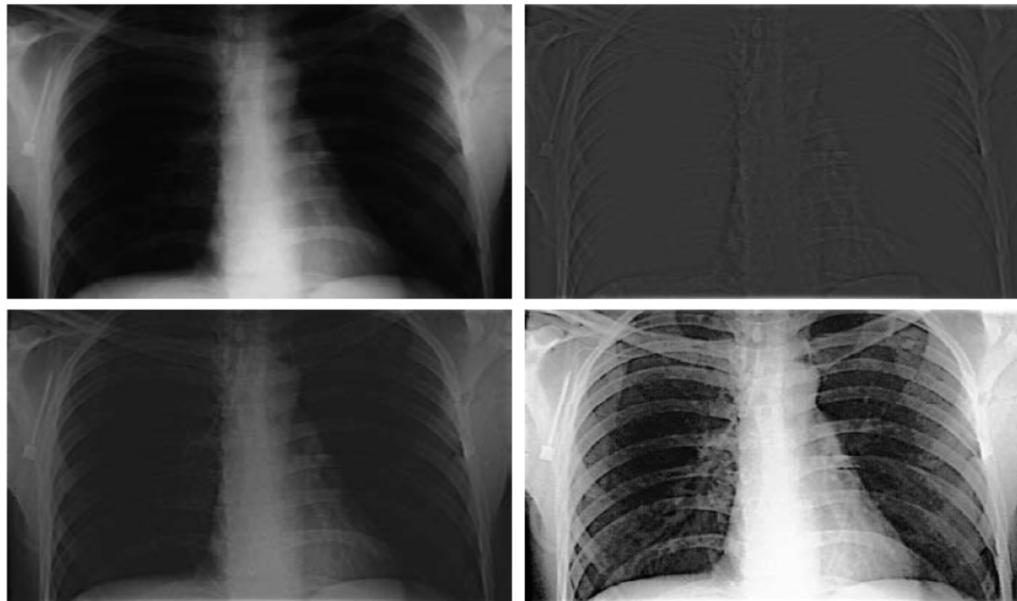
$$g(x,y) = F^{-1}\{[1 + k * [1 - H_{lp}(u,v)]]F(u,v)\}$$

- 或 $g(x,y) = F^{-1}\{[1 + k * H_{hp}(u,v)]F(u,v)\}$

高频强调滤波器

高频强调滤波例子

- 更一般形式: $g(x,y) = F^{-1} \{ [k_1 + k_2 * H_{hp}(u,v)] F(u,v) \}$



a b
c d

FIGURE 4.59 (a) A chest X-ray image. (b) Result of highpass filtering with a Gaussian filter. (c) Result of high-frequency-emphasis filtering using the same filter. (d) Result of performing histogram equalization on (c). (Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

- $K1=0.5, K2=0.75$
- 直方图均衡
- 医学图像处理中，振铃现象是不可接受的，采用高斯滤波器可以避免振铃。

原图	高斯高通滤波结果
高斯高频强调滤波	对c进行直方图均衡结果

同态滤波

- 先对图像作对数运算，即变换到对数亮度域中——称为密度域。密度域处理更符合人类视觉、听觉特性！

- 同态图像滤波 $f(x,y) \Rightarrow \boxed{\text{对数}} \Rightarrow \boxed{\text{增强处理}} \Rightarrow \boxed{\text{指数}} \Rightarrow g(x,y)$

- 图像的形成: $f(x,y) = i(x,y) \cdot r(x,y)$

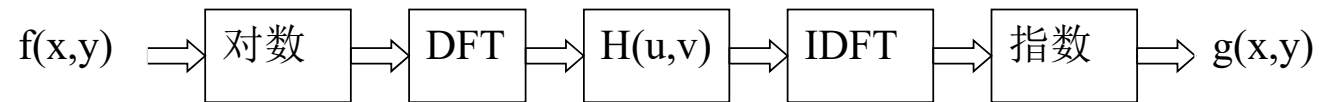
- 同态分析: $z(x,y) = \ln f(x,y) = \ln i(x,y) + \ln r(x,y)$

然后可采用以前讨论的各种增强方法对 $z(x,y)$ 进行增强处理，最后取指数

同态图像滤波

- 图像的照射分量通常由慢的空间变化来表征，而反射分量往往引起突变，特别是在不同物体的连接部分。
- 这些特性导致图像取对数后的傅里叶变换的**低频成分与照射相联系**，而**高频成分与反射相联系（近似地）**，因而可以采用不同方法衰减（增强）照射（反射）的贡献。

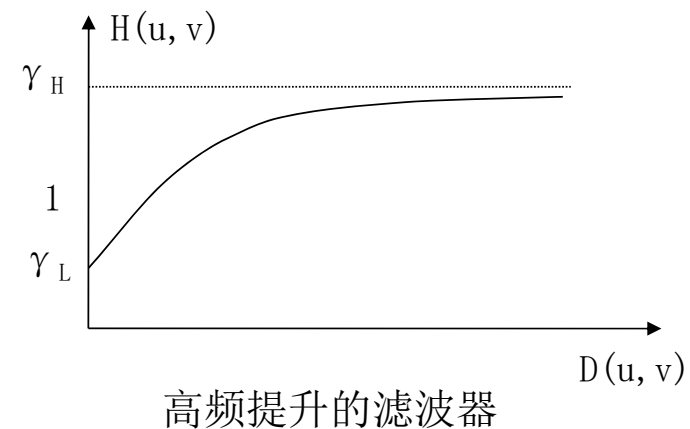
例：同态处理



其中

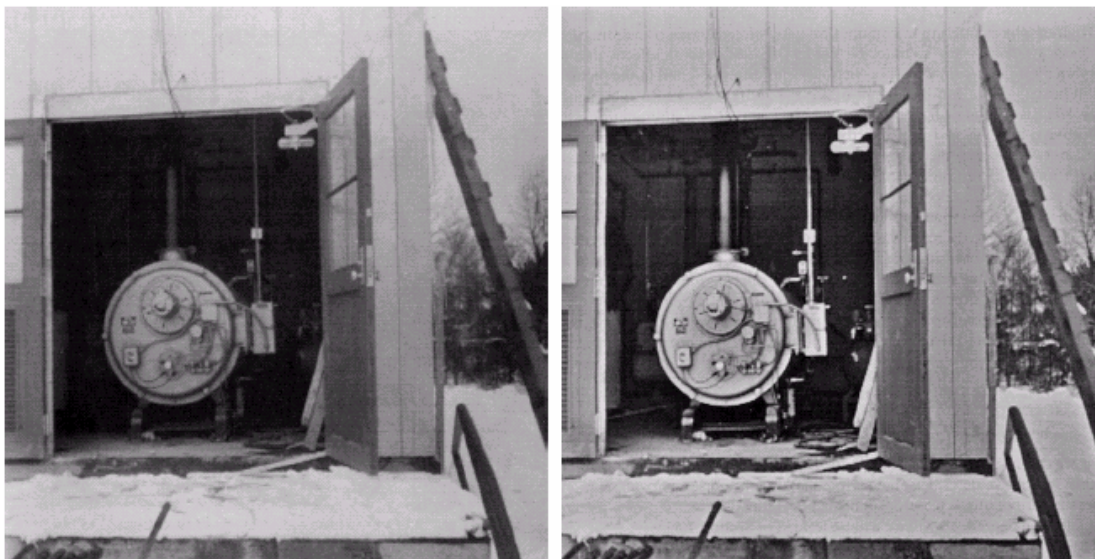
$$H(u,v) = (\gamma_H - \gamma_L) \left[1 - e^{-C \frac{D^2(u,v)}{D_0^2}} \right] + \gamma_L$$
$$D(u,v) = (u^2 + v^2)^{\frac{1}{2}}$$

C用于控制滤波器的陡峭程度
若 $\gamma_L < 1$ 且 $\gamma_H > 1$ ，则衰减低频（照射），增强高频（反射），同时进行动态范围的压缩和对比度的增强。

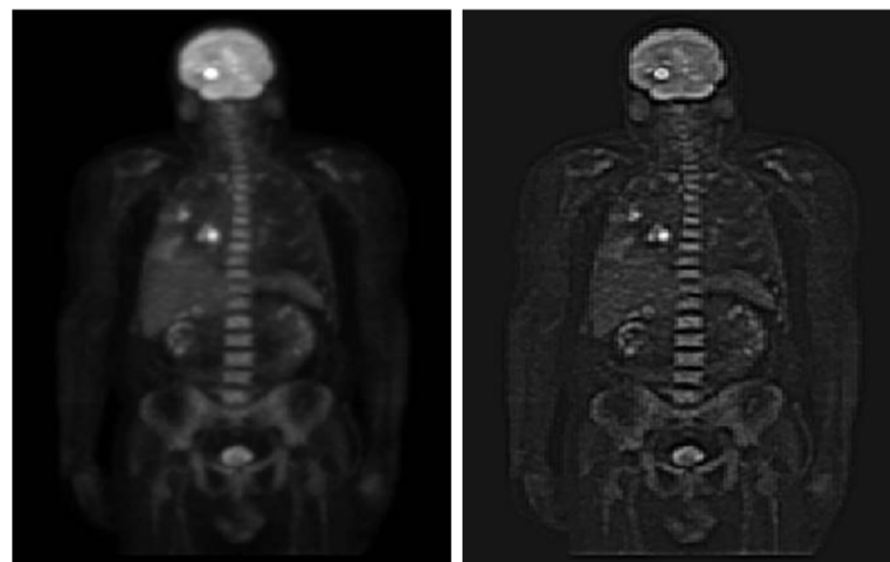


同态滤波的特点:

- (1) 由于需要进行对数和指数运算, 计算量大。
- (2) 对于加性噪声, 不能用同态滤波滤除。



同态图像滤波: (a) 原图, (b) 处理结果 $\gamma_H=2.0$, $\gamma_L=0.5$



(a) PET 扫描图像 (b) 同态滤波增强的图像

选择性滤波

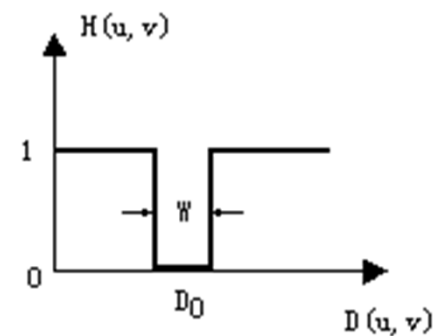
带阻滤波器

理想带阻

$$H(u, v) = \begin{cases} 0 & \text{若 } D_0 - W/2 \leq D(u, v) \leq D_0 + W/2 \\ 1 & \text{其它} \end{cases}$$

其中 $D(u, v) = (u^2 + v^2)^{\frac{1}{2}}$

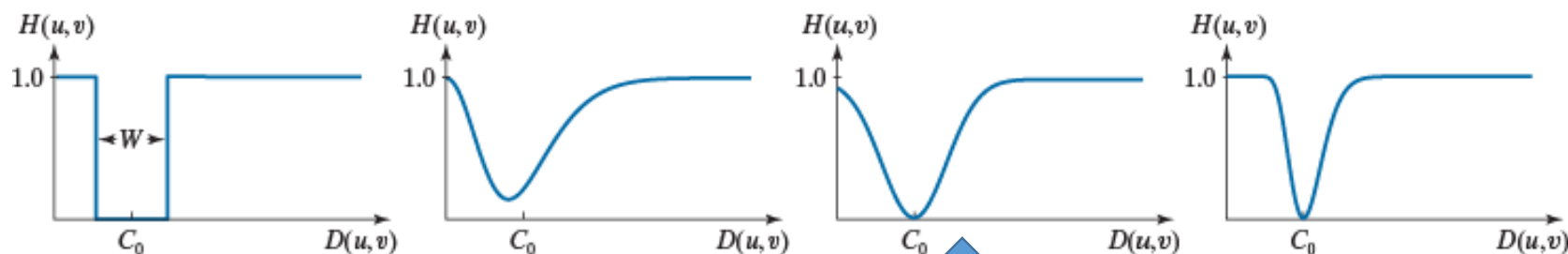
W 为阻带宽度, D_0 为阻带中心频率



理想带阻滤波器

高斯带阻滤波器

$$H(u, v) = 1 - e^{-\frac{1}{2} \left[\frac{D^2(u, v) - D_0^2}{D(u, v)W} \right]^2}$$



理想带阻滤波器

高斯低通+高通滤波器

$$H(u, v) = 1 - e^{-\frac{[(D(u, v) - C_0)^2]}{W^2}}$$

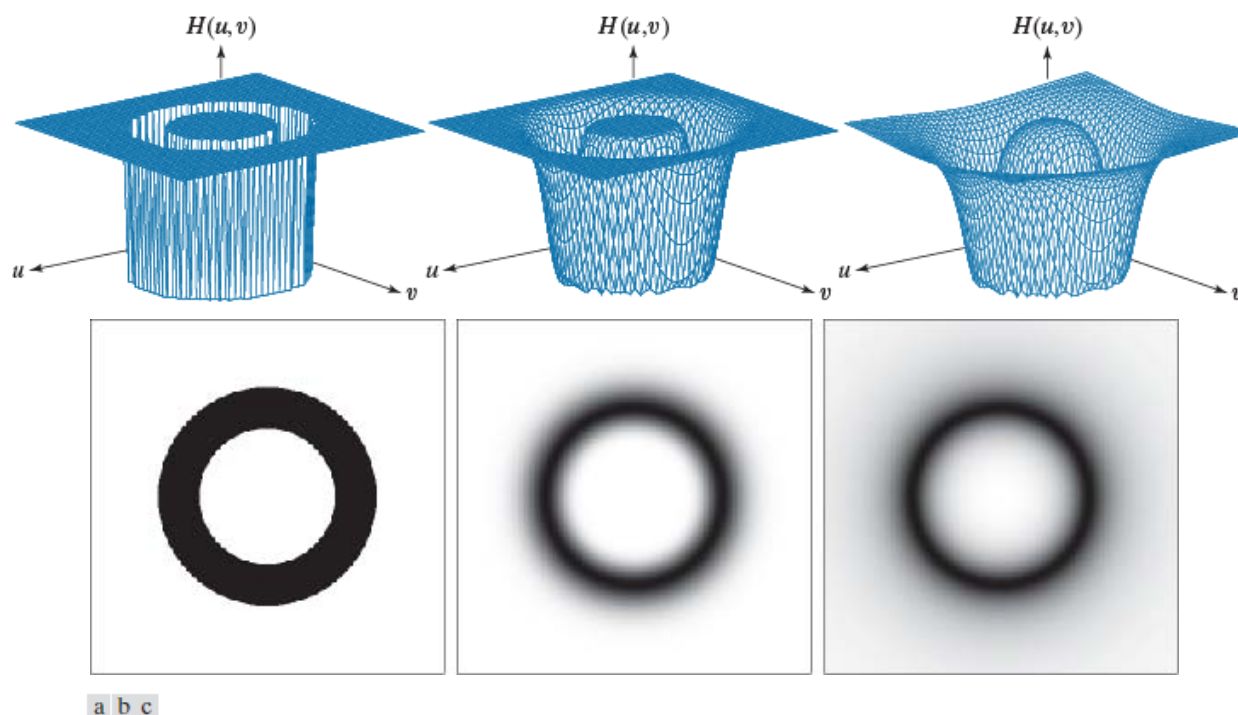
$$H(u, v) = 1 - e^{-\frac{1}{2} \left[\frac{D^2(u, v) - D_0^2}{D(u, v)W} \right]^2}$$

帶阻濾波器

TABLE 4.7

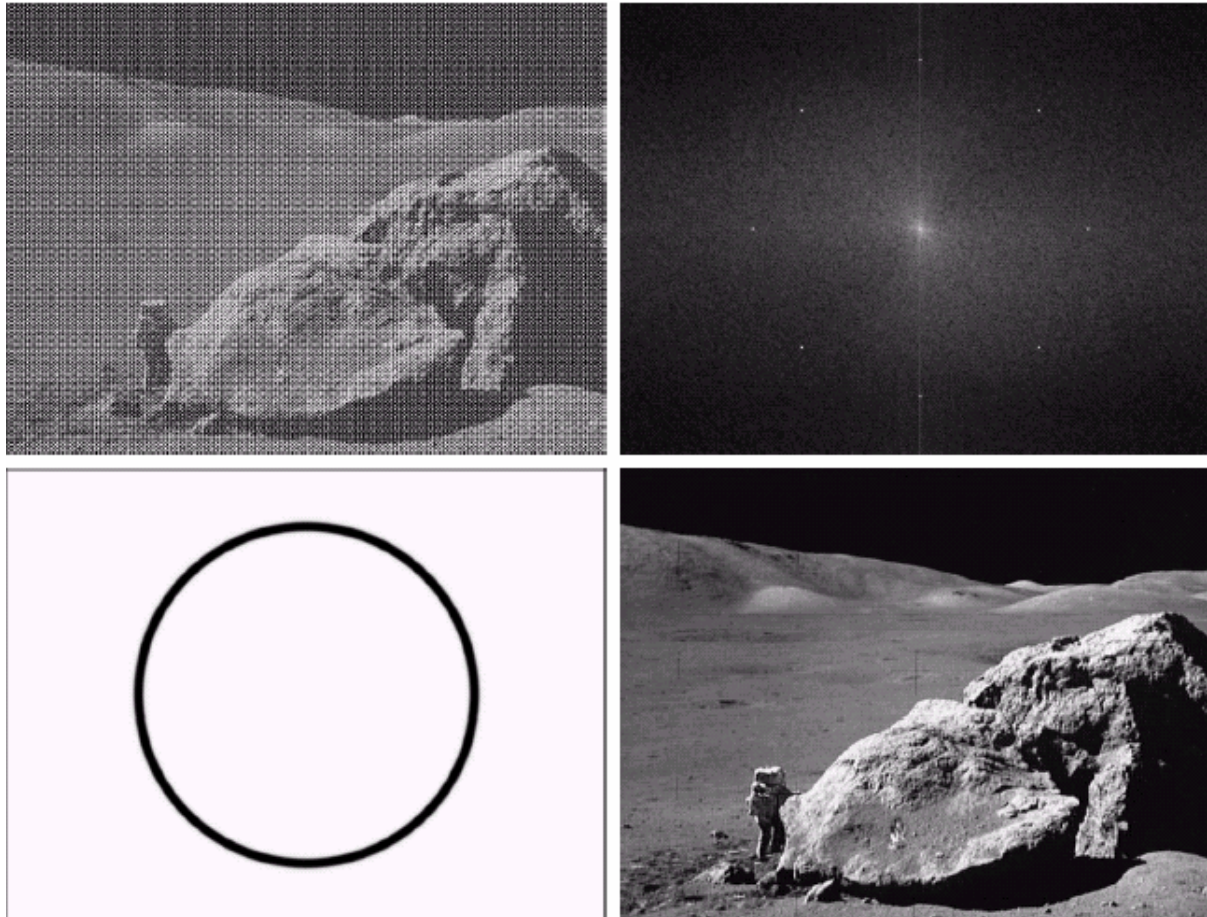
Bandreject filter transfer functions. C_0 is the center of the band, W is the width of the band, and $D(u,v)$ is the distance from the center of the transfer function to a point (u,v) in the frequency rectangle.

Ideal (IBRF)	Gaussian (GBRF)	Butterworth (BBRF)
$H(u,v) = \begin{cases} 0 & \text{if } C_0 - \frac{W}{2} \leq D(u,v) \leq C_0 + \frac{W}{2} \\ 1 & \text{otherwise} \end{cases}$	$H(u,v) = 1 - e^{-\left[\frac{D^2(u,v) - C_0^2}{D(u,v)W}\right]^2}$	$H(u,v) = \frac{1}{1 + \left[\frac{D(u,v)W}{D^2(u,v) - C_0^2}\right]^{2n}}$



a b c

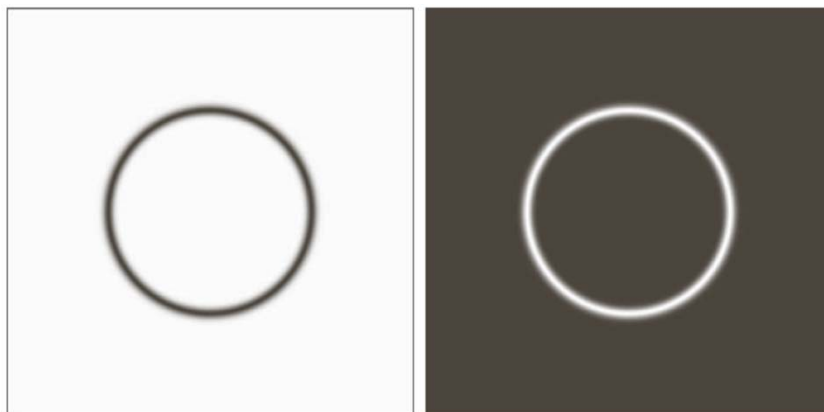
FIGURE 4.62 Perspective plots of (a) ideal, (b) modified Gaussian, and (c) modified Butterworth (of order 1) bandreject filter transfer functions from Table 4.7. All transfer functions are of size 512×512 elements, with $C_0 = 128$ and $W = 60$.



(a)被正弦噪声污染的图像, (b)图(a)的频谱图,
(c) 4阶巴特沃思带阻滤波器(白代表1), (d)滤波结果

帶通濾波器

$$H_{bp}(u, v) = 1 - H_{br}(u, v)$$



高斯帶阻和帶通濾波器

陷波滤波器

消除以 (u_0, v_0) 为中心, D_0 为半径的区域内的所有频率分量

$$H(u, v) = \begin{cases} 0 & \text{若 } D(u, v) \leq D_0 \\ 1 & \text{若 } D(u, v) > D_0 \end{cases}$$

其中 $D(u, v) = [(u - u_0)^2 + (v - v_0)^2]^{1/2}$

$$H(u, v) = \begin{cases} 0 & \text{若 } D_1(u, v) \leq D_0 \text{ 或 } D_2(u, v) \leq D_0 \\ 1 & \text{其它} \end{cases}$$

$$D_1(u, v) = [(u - u_0)^2 + (v - v_0)^2]^{1/2}$$

$$D_2(u, v) = [(u + u_0)^2 + (v + v_0)^2]^{1/2}$$

- 零相位陷波滤波器须对原点对称

陷波滤波器

- 若考虑多个陷波滤波器，则

$$H_{NR}(u,v) = \prod_{k=1}^Q H_k(u,v) H_{-k}(u,v)$$

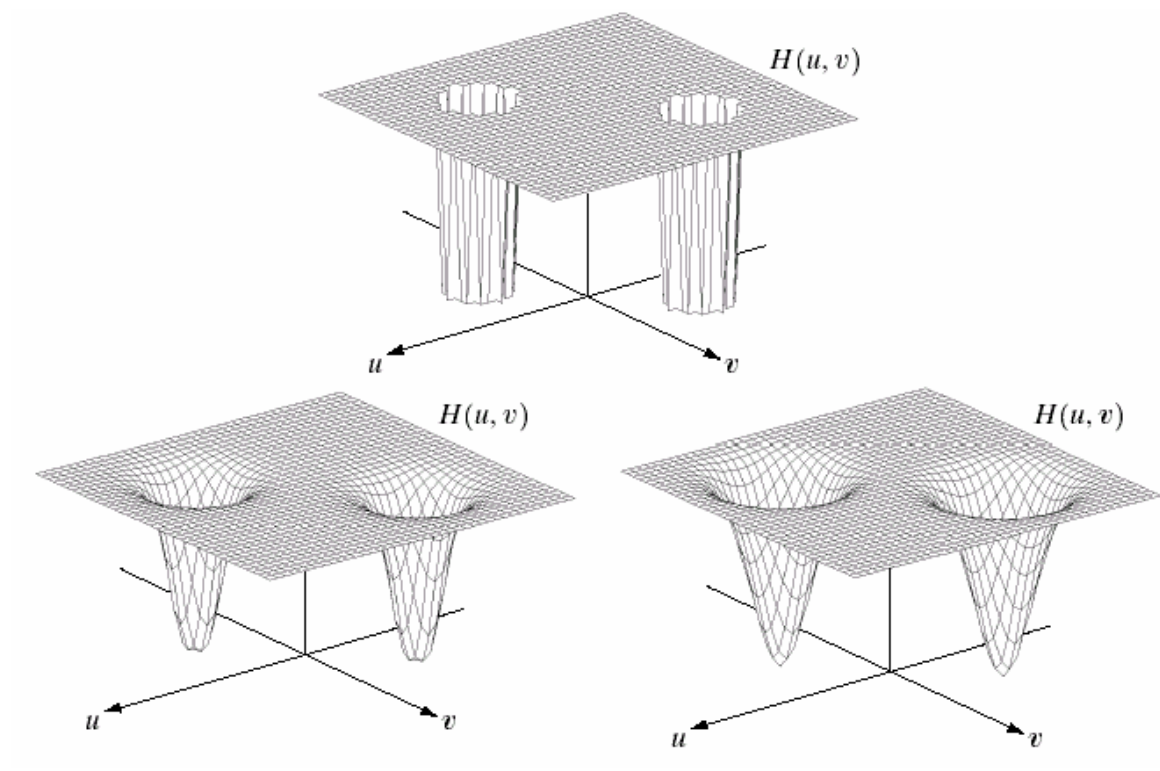
- 其中

$$D_k(u,v) = [(u - u_k)^2 + (v - v_k)^2]^{1/2}$$

$$D_{-k}(u,v) = [(u + u_k)^2 + (v + v_k)^2]^{1/2}$$

例：含三个陷波对的巴特沃思型陷波滤波器

$$H_{NR}(u,v) = \prod_{k=1}^3 \left[\frac{1}{1 + [D_{0k} / D_k(u,v)]^{2n}} \right] \left[\frac{1}{1 + [D_{0k} / D_{-k}(u,v)]^{2n}} \right]$$

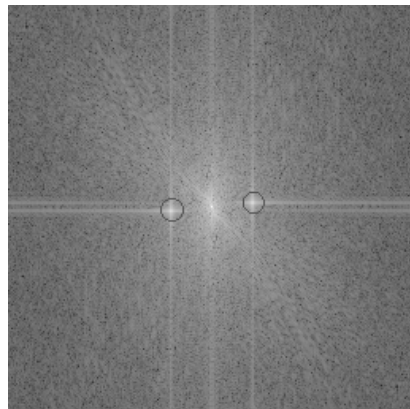


陷波滤波器：（a）理想型（b）巴特沃思型
（c）高斯型

陷波滤波器实例



(a) 受条纹干扰的图像

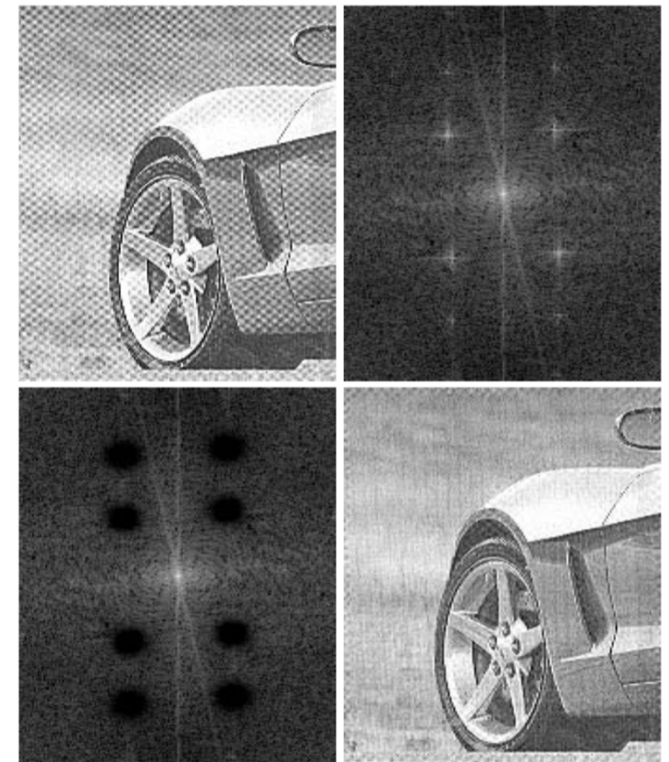


(b) 频谱图



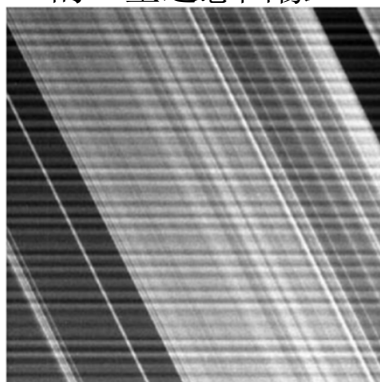
(c) 陷波滤波后的图

正弦型干扰及其理想陷波滤波

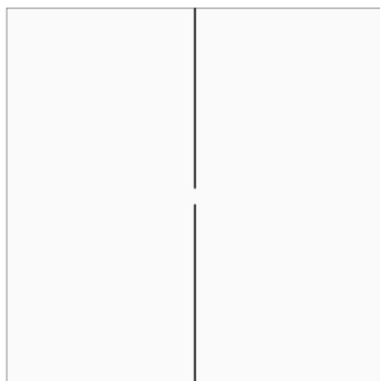
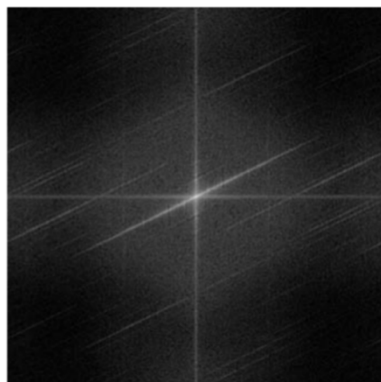


陷波滤波器实例

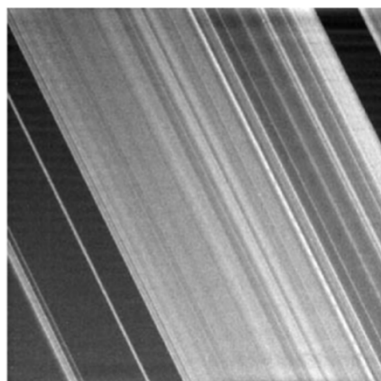
(a) 有水平条纹干扰的卫星遥感图像



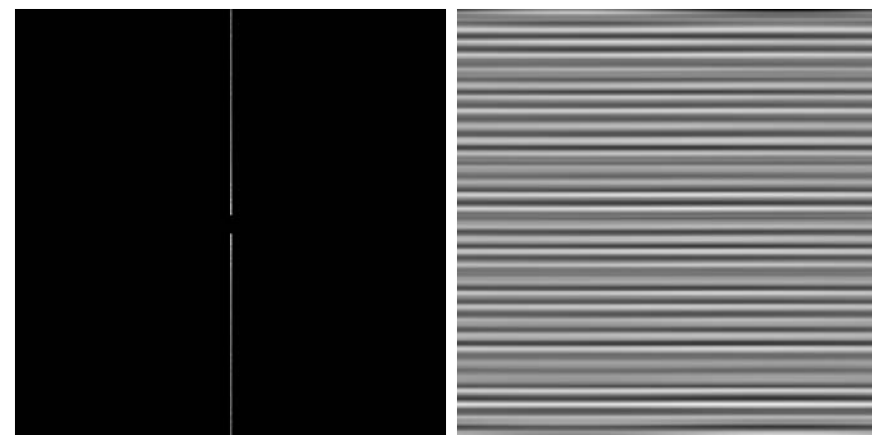
(b) 图 (a) 的频谱图



(c) 陷波滤波器(白色为, 黑色为0)



(d) 陷波滤波后的图



左：陷波带通滤波器(白色为, 黑色为0),
(右) 获得的周期干扰图

FIGURE 4.66
(a) Result (spectrum) of applying a notch pass filter to the DFT of Fig. 4.65(a).
(b) Spatial pattern obtained by computing the IDFT of (a).

离散傅里叶变换的实现

- 二维DFT的可分性

$$F(u,v) = \sum_{x=0}^{M-1} e^{-j2\pi ux/M} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi vy/N} = \sum_{x=0}^{M-1} F(x,v) e^{-j2\pi ux/M}$$

- 其中：
$$F(x,v) = \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi vy/N}$$

- 二维DFT可通过两个依次按照行和列的一维DFT变换来得到，这是一个重要的简化。
- 先计算每一行的一维DFT，再沿着计算结果的每一列进行一维DFT

用DFT算法计算IDFT

- 由 $MNf^*(x,y) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} F^*(u,v) e^{-j2\pi(ux+vy)/N}$ 可知，如果将 $F^*(u,v)$

代入DFT，将获得 $MNf^*(x,y)$ 。由此容易得到真正需要的 $f(x,y)$

快速Fourier变换 (FFT)

- 一. FFT算法原理
 - DFT正变换

$$F(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) W_N^{ux}, \quad u=0,1,\dots,N-1$$

其中

$$W_N = e^{-j2\pi/N}$$

DFT计算每个**F(u)**都要做**N**次复数乘法和**N-1**次加法,共要进行**N*N**次乘法和**N*(N-1)**次加法

• FFT导出

- 设**N**为**2**的正整数次幂:

$$N = 2^n$$

- 设**M**为一正整数, 使

$$N = 2M$$

- 则有:

$$F(u) = \frac{1}{2M} \sum_{x=0}^{2M-1} f(x) W_{2M}^{ux} = \frac{1}{2} \left[\frac{1}{M} \sum_{x=0}^{M-1} f(2x) W_{2M}^{u(2x)} + \frac{1}{M} \sum_{x=0}^{M-1} f(2x+1) W_{2M}^{u(2x+1)} \right]$$

- 由于 $W_{2M}^{2ux} = e^{-(j2\pi/2M)2ux} = e^{-(j2\pi/M)ux} = W_M^{ux}$

- 故 $F(u) = \frac{1}{2} \left[\frac{1}{M} \sum_{x=0}^{M-1} f(2x) W_M^{ux} + \frac{1}{M} \sum_{x=0}^{M-1} f(2x+1) W_M^{ux} W_{2M}^u \right]$

- 定义:
$$F_{\text{even}}(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(2x) W_M^{ux}, \quad u=0,1,\dots,M-1$$

$$F_{\text{odd}}(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(2x+1) W_M^{ux}, \quad u=0,1,\dots,M-1$$

- 则有:
$$F(u) = \frac{1}{2} [F_{\text{even}}(u) + F_{\text{odd}}(u) W_{2M}^u]$$

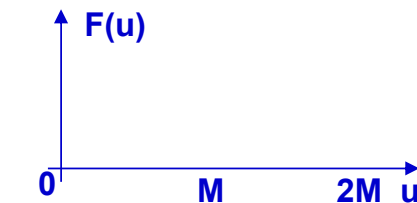
- 由于 $W_M^{u+M} = W_M^u$ 和 $W_{2M}^{u+M} = -W_{2M}^u$

- 代入上式, 得
$$F(u+M) = \frac{1}{2} [F_{\text{even}}(u) - F_{\text{odd}}(u) W_{2M}^u]$$

- 设 N ($N = 2^n$) 点 **FFT** 需要的乘法和加法次数分别为 $m(n)$ 和 $a(n)$, 则

$$m(n) = 2m(n-1) + 2^{n-1} \quad n \geq 1$$

$$a(n) = 2a(n-1) + 2^n \quad n \geq 1$$



*两个一半长度的**DFT**序列计算, 外加一半数目的乘法(另一半可共用), 和全部数目的加法

- 由于 **$n=0$** ,时 **$2^n=1$** ,单点的**DFT**为其本身, $m(0) = 0, a(0) = 0$

- 故
$$m(n) = \frac{1}{2} 2^n \log_2(2^n) = \frac{1}{2} N \log_2 N \quad n \geq 1$$

$$a(n) = 2^n \log_2(2^n) = N \log_2 N \quad n \geq 1$$

- 计算量的减少:

- 如 **$N=1024$** , 乘法次数的减少量:
即只有原来的约**1/200**

$$\frac{N^2}{\frac{1}{2} N \log_2 N} = \frac{2N}{\log_2 N} = 204.8$$

- 加法次数只有原来的约**1/100**

• 二. 快速Fourier反变换 (IFFT)

- $$F^{-1}\{F(u)\} = f(x) = \sum_{u=0}^{N-1} F(u) \exp[j2\pi ux / N] \quad , \quad \mathbf{x=0,1,...,N-1}$$

两边取共轭

$$f^*(x) = N \times \frac{1}{N} \sum_{u=0}^{N-1} F^*(u) \exp[-j2\pi ux / N]$$

• 对于二维情况

$$f(x, y) = \frac{1}{N} \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} F(u, v) \exp[j2\pi(ux + vy) / N]$$

, x,y=0,1,...,N-1

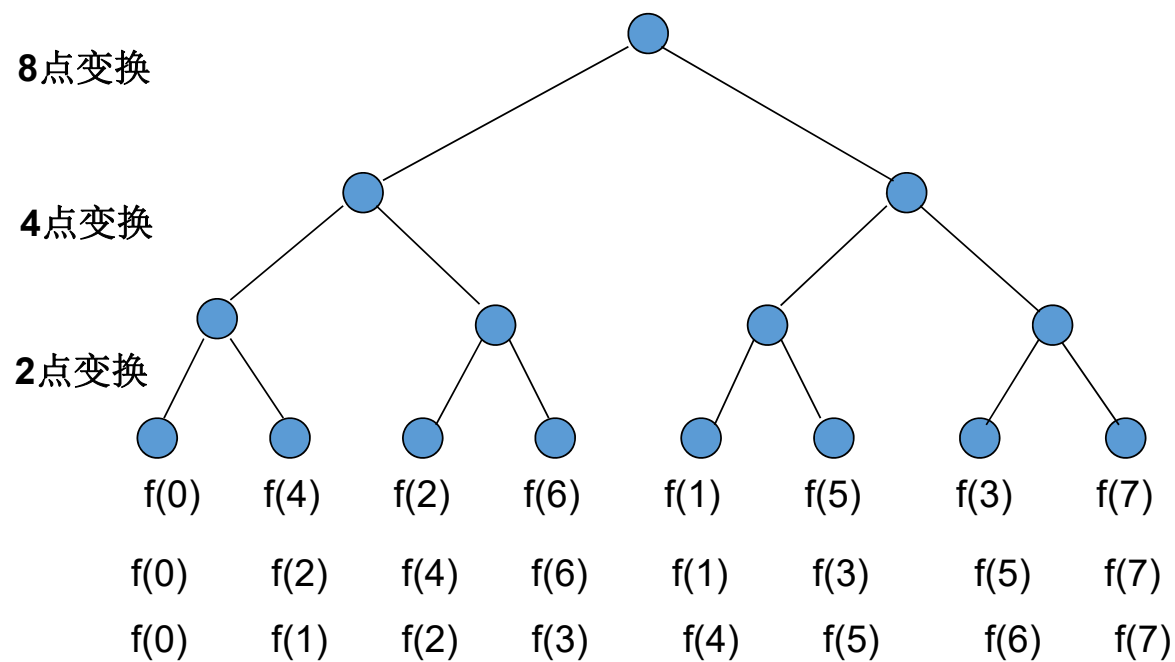
$$f^*(x, y) = \frac{1}{N} \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} F^*(u, v) \exp[-j2\pi(ux + vy) / N]$$

• 而对图像

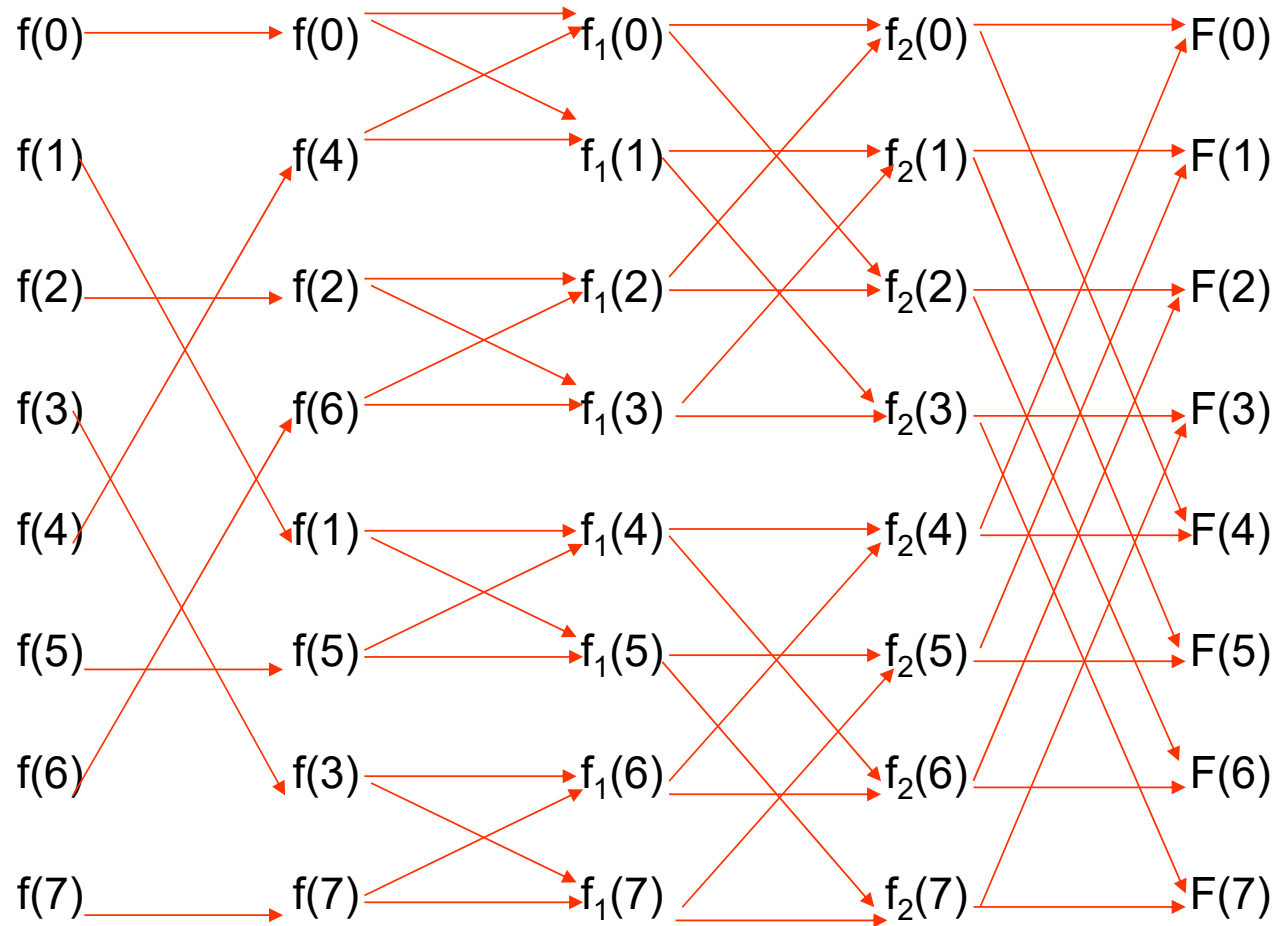
$$f(x, y) = f^*(x, y)$$

• 三. FFT算法的实现

- 先对输入进行排序:可用二进制序号高低反转进行定序
 - 如, $6 = (110)_2$, 倒序后为 $(011)_2 = 3$, 故排在第3位



FFT蝶形运算流程



课后作业

- 习题4.10, 4.26, 4.41, 4.42, 4.43, 4.44, 4.47
- 上机作业（二选一）：
 1. 实现一个 2D图像的FFT算法，并将频谱移到图像中心；
 2. 实现一个傅里叶频率域滤波方法，对一幅图像进行高频提升滤波。